The Spatial Cost of Formality: Street Vendors' Preferences in India

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Abstract

Formalisation policies in developing countries have had mixed success, and their effects depend on design. Firms that operate on public land pose a distinct problem: the state cannot grant them legal status without also choosing where they may trade, so formality is offered at a location rather than at a price. We study this trade-off for street vendors, who account for 1.5 to 6.8% of employment in low- and middle-income countries and are commonly formalised through government-built vending zones. We embed a stated-preference experiment in an original geolocated survey of 2,922 vendors in 206 marketplaces in Patna, India, where the Street Vendors Act of 2014 makes certification free but ties legal protection to a vending zone. At an unchanged commute, the average vendor's willingness to pay for a protected spot is 4.5% of monthly sales, and predicted take-up of a free spot is 68%. Willingness to pay falls to 2.1% of sales when the commute rises by 1 km and reaches zero at about 3.8 km; a 5 km move reduces take-up twice as much as a rent of 2% of sales. A randomised information treatment raises take-up by about 3 percentage points at every design. The results indicate that low take-up of vending zones reflects the relocation cost that place-based formalisation imposes, not low demand for formal status.

JEL Classification: O17, R38, C25, O18, C93
Keywords: Informality, Firm Location, Relocation costs, Vending Zones, Stated Preferences

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1 Introduction

Around 2.1 billion people, or 58 per cent of the global workforce, work in the informal sector (ILO 2026). Governments in developing countries often attribute low fiscal capacity and low aggregate productivity to persistent informality. Many have responded with policies that grant informal firms legal rights and social protection upon registration. Registration rates nonetheless remain low. How much a given policy achieves depends on its design and on how much the targeted firms value what it offers.

Street vendors fit awkwardly into this framework. They are own-account firms that sell goods and services from carts, stalls and ground displays in public spaces to which they have no legal right. The International Labour Organization's definition covers vendors who trade from a fixed pitch and those who trade while mobile (ILO 2013). Street vending accounts for 1.5 to 6.8 per cent of total employment in low- and middle-income countries (Roever and Skinner 2016). A government cannot confer legal status on these firms without also deciding where they may lawfully trade. It therefore offers formality at a location, not at a price, and accepting it means relocating the firm.

Where vendors trade is contested. They concentrate on main roads and pavements that vehicles and pedestrians also claim. Since 2010, governments have legislated extensively to ease congestion, keep public spaces usable and bring vendors onto a register. Examples include Accra's decongestion by-laws, Bangkok's Return the Footpath to Pedestrians, Ho Chi Minh City's Civilized Streets and Johannesburg's Operation Clean Sweep. Appendix A1 documents the non-exhaustive list of 54 such policies in 27 countries across six regions. Just over 70 per cent of them designate or construct a physical site to which vendors must move. Typically, a demarcated, legally protected area in which certified vendors receive fixed-size spots for a monthly rent. Vendors receive the protected spot only if they relocate, and the standard instruments of formalisation policy do not address this condition.

Those instruments have been studied closely. The administrative and monetary costs of becoming and remaining formal are well documented. Most of the evidence finds that removing the fixed costs of registration raises formality by less than reducing the recurring costs of formality or raising the cost of informality through enforcement (Bruhn and McKenzie 2014; Rocha, Ulyssea, and Rachter 2018; Farias and Hsu Rocha 2026; Andrade, Bruhn, and McKenzie 2016; Ulyssea 2020; Campos, Goldstein, and McKenzie 2023), although which margin binds remains contested. Far less work examines the relocation costs that formality can carry, or firms operating on public land. Where land appears at all, it appears as an eligibility condition, not as a price. de Mel, McKenzie, and Woodruff (2013) offered informal firms in Sri Lanka payments worth up to two months of median profits to register. Firms operating on public land were 27 percentage points less likely to register, because they could not produce documents establishing a claim to their premises. Firms screened out on documentation never face the choice, so we never observe what formality is worth to them or what they would give up for it.

This paper measures the relocation costs that formalisation imposes on firms without premises, and the preferences of the firms the policy targets. We separately identify and value four potential barriers to relocation: commuting and location costs, rent, coordination failure among vendors, and limited information about the policy. We also test a fifth

design margin, the number of spots in a zone. This lets us identify the binding constraint and show how much scope redesign has. In doing so, we create new and detailed data on a group of informal firms about which little is known.

India's Street Vendors (Protection of Livelihood and Regulation of Street Vending) Act, 2014 provides a useful setting. It requires every urban local body to survey and certify vendors and to establish vending zones. Certification is free, is issued upon survey, and does not require proof of a claim to the vendor's current site. The Act therefore removes the fixed registration costs at the centre of much of the formalisation literature. Certification does not, however, secure the vendor's current pitch. The protection it brings attaches to a spot inside a vending zone, and certified vendors in our sample report more harassment and eviction than uncertified ones (Appendix A2). Low take-up therefore points to the terms of relocation, not to the cost of registering. We study the Act's implementation in Patna, the capital of Bihar. About 30,000 vendors operate in Patna's public spaces, and the Municipal Corporation has built several vending zones, all but one of which have zero occupancy.

Because vendors rarely change marketplaces, revealed preference cannot tell us how vendors decide where to relocate. The zones are too few and too empty for their attributes to vary in the field. And the share of peers who relocate is determined jointly with each vendor's own decision. We therefore use a stated-preference experiment, which recovers valuations of attributes that do not vary in the field (McFadden 1974). We conducted an original geolocated survey of 2,922 vendors across 206 marketplaces in Patna, recording each vendor's location, rent, commute and trading conditions. Each vendor then chooses five times between staying at the current site and moving to a hypothetical vending zone. The zone's distance, rent and expected trading conditions are set relative to the vendor's verified status quo. Between rounds, about half the vendors in every marketplace receive information about the Act and the credibility of its enforcement. Section 3 describes the design, the discrete-choice model and the steps we take to limit hypothetical bias.

Vendors value formality. Suppose a municipality designates an existing site as a zone, so that no commute changes. The average vendor's willingness to pay for the protected spot is then 4.5 per cent of monthly sales, and predicted take-up of a free spot is 68 per cent. Vendors value protection, not productivity. This accords with evidence that registration raises neither earnings nor productivity for most firms, and that its benefits come from reduced risk (de Mel, McKenzie, and Woodruff 2013; Benhassine et al. 2018; Farias and Hsu Rocha 2026; Ulyssea 2020).

Relocation erodes that value. Willingness to pay falls to 2.1 per cent of sales after a commute one kilometre longer, and reaches zero at about 3.8 kilometres. Beyond that distance, the average vendor refuses the zone even when the spot is free. The cost reflects what the vendor gives up by leaving the site. Before seeing any choice, the median vendor expects only 5 per cent of regular customers to follow to a new site, so the customer base does not move with the firm.

Information about the policy and evidence of its enforcement raise the value of relocating. The treatment lifts predicted take-up by about 3 percentage points at every design we evaluate, and we cannot reject that it shifts only the level of the valuation, leaving the

trade-offs between distance, rent and the zone's other attributes unchanged (Section 3.3). A gain of this size falls well short of the 22 points a 5 km move costs, so information cannot make up for a distant zone.

The remaining explanations account for less. Rent carries disutility from the first percentage point of sales, but a 5 km move lowers take-up by 22 percentage points, twice the 11 points lost by charging 2 per cent of sales at an unchanged commute. We find no detectable response to randomised variation in the share of peers a vendor expects to relocate, although coordination failure is the explanation most commonly advanced for unoccupied formal space (Cooper and John 1988; Brock and Durlauf 2001). The design limits how far this null extends: the shock varies an anonymous, stated share over a censored range, so it rules out a large response to that share, not coordination failure in every form. The number of spots a zone offers has no detectable effect, so building larger zones is not a remedy. One margin we cannot price is the size of each spot. Our choice tasks tell vendors that the zone gives them the same storage space they have now. Stated willingness to relocate may therefore overstate take-up of the zones as built.

Vendors differ widely in how much they value a zone, and most of this variation is unobserved. We control for 27 vendor characteristics, including earnings, education, tenure and exposure to eviction, yet together they account for only 15.5 per cent of the variation in valuations (Section 4.1.2). Targeting zones at vendors on observable characteristics is therefore unlikely to raise take-up much. What a municipality can vary instead is the terms of the offer.

These results contribute to the formalisation and relocation literatures. To our knowledge, this is the first paper to value the barriers to formalisation facing firms without premises on a common scale, each vendor's monthly sales. This lets us rank the barriers instead of only listing them. Distance is the binding constraint: a long move costs far more take-up than a moderate rent, and information has a modest effect that leaves the cost of distance unchanged. The paper also prices location, which the formalisation literature generally treats only as an eligibility condition. The Act removes the documentary requirements that would otherwise exclude firms without premises. We can therefore directly estimate the cost of exchanging location for legal status, and we observe the preferences of firms that such requirements would ordinarily screen out before any choice is made.

The relocation literature has mainly studied households. Evaluations of slum relocation find that destination location matters as much as housing quality, and that a better dwelling in a worse location may leave households no better off (Takeuchi, Cropper, and Bento 2008; Barnhardt, Field, and Pande 2017). The slum-upgrading literature reaches the same conclusion from the other direction, showing how much of the long-run gain from secure tenure depends on where it is conferred (Harari and Wong 2026). The evidence is mixed, however. In Addis Ababa, subsidised relocation to peripheral housing had high take-up and no adverse labour-market effects (Franklin 2025; Agness and Getahun 2024). Discrete-choice studies of what relocation sites must offer have relied on moves that already occurred (Takeuchi, Cropper, and Bento 2008; Lall, Lundberg, and Shalizi 2008). For households, location is partly a consumption choice, and movers can change employers to offset some losses. For firms, location is an input into production, and no

comparable substitution is available. Because we assign destination attributes experimentally, we can trace how the cost of relocation grows with distance and locate the distance at which it exhausts the value of formality.

The results limit what policy design can achieve. Where a municipality can designate an existing vending site as a zone, vendors receive the protected spot without incurring the cost of moving. On our estimates, this is the best option. It is also the option the municipality has least room to choose, because the sites vendors occupy are the sites whose congestion the policy exists to relieve. Where relocation is unavoidable, waiving rent cannot offset a long move: a 5 km move costs twice the take-up that a rent of 2 per cent of sales does.

The remainder of the paper is organised as follows. Section 2 describes the institutional setting, sampling frame and survey. Section 3 presents the stated-preference design, the discrete-choice model and the information experiment. Section 4 reports the results, and Section 5 discusses their implications and limitations.

2 Data

2.1 Institutional Context

In India, the Street Vendors (Protection of Livelihood and Regulation of Street Vending) Act, 2014 (SVA henceforth) governs street vending. The Act requires every urban local body to survey the street vendors in its jurisdiction, issue certificates of vending, and designate vending zones to which it allocates vendors. State governments frame the rules and schemes under which local bodies discharge these obligations. Vendors accommodated in a zone receive legal recognition and protection from eviction, rights they lack at their current informal locations. Recognition under the SVA also makes them eligible for complementary government schemes, most notably the Pradhan Mantri Street Vendor's AtmaNirbhar Nidhi (PM SVANidhi), which provides subsidised collateral-free loans. Town Vending Committees (TVCs) oversee implementation at the municipal level, and state governments retain discretion over licensing terms, fee structures and renewal conditions. As a result, implementation varies substantially across cities.

The Act's core mechanism is spatial: it removes some costs of formality and relocates others. Certification carries no fee, and the vendor files nothing. The municipality conducts the survey at the marketplace and registers vendors there. Eligibility also does not rest on documenting a claim to the site occupied, a requirement that excludes firms trading on public land elsewhere. The state, not the firm, therefore bears the fixed cost of entering the formal sector, on which much of the formalisation literature has focused. What remains are recurring obligations. The scheme framed under the Act provides for a monthly charge for a spot inside a vending zone, and vendors must renew their certificate of vending every five years for a small fee (India 2014). Legal status is therefore conditioned neither on a payment at entry nor on documented tenure. It is conditioned on where the vendor trades.

Certification on its own does not protect the site a vendor occupies. Twenty per cent of our sample hold a certificate of vending, and they report more enforcement exposure

than uncertified vendors rather than less: 61 against 38 per cent report harassment by officials or the police in the preceding six months, and 46 against 30 per cent report an eviction or a confiscation of goods (Appendix Table A2). Marketplace fixed effects cut these gaps by roughly a third, so part of the difference is which marketplaces PMC chose to survey; the most congested marketplaces are both the likeliest to be surveyed and the likeliest to be cleared. Most of the gap, however, is between vendors trading in the same marketplace, and certified vendors are no more likely than uncertified ones to be in a marketplace with frequent eviction drives. At the vendor level the more plausible reading is the reverse of protection: vendors who have been harassed or evicted are the ones who seek a certificate, and the certificate does not change what happens at the pitch they go on occupying. Either way the implication is the same. The right the Act confers is a protected spot inside a vending zone, so until a vendor takes one up the certificate carries no spatial protection at all.

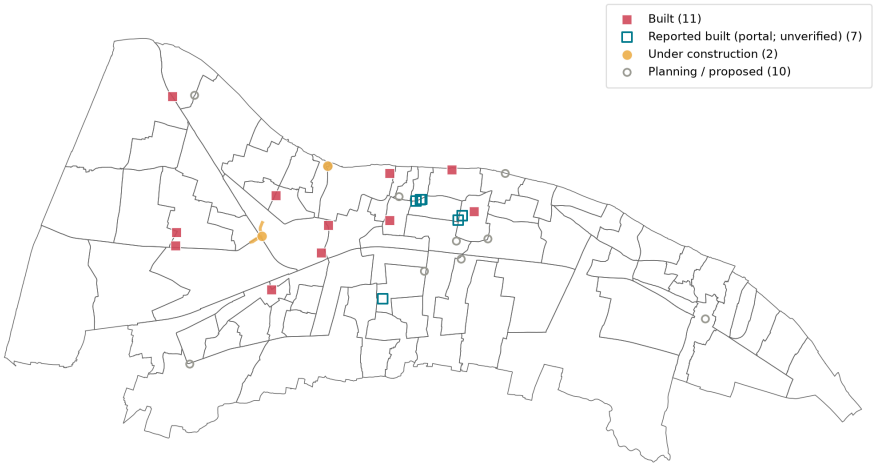
The siting of zones is a central policy decision. Policymakers state that they relocate vendors to relieve congestion on carriageways and pedestrian walkways, so the sites where vendors concentrate are precisely the sites a municipality seeks to clear. The same density constrains the alternatives: a zone requires a plot of a given size, and the built-up centre of a congested city has few such plots. Both forces push zones away from the corridors on which vendors currently trade, and they constrain the offer a municipality implementing the Act can make.

We chose Patna, the capital of Bihar, as the study site for three reasons. First, a single statutory authority, the Patna Municipal Corporation (PMC), governs its entire urban area of 75 wards. One set of rules covers all street vendor regulation, including licensing, enforcement and policy implementation. This unified jurisdiction removes the confounding effect of inter-agency variation present in cities such as Delhi, where several civic bodies govern different areas and may enforce differently. It also simplifies both data collection and the interpretation of vendor behaviour.

Second, commuter-led demand, not tourism, shapes Patna's street vending. Vendors serve a stable local customer base of office workers, daily-wage earners and households. They sell necessities, mainly produce and food for immediate consumption, at prices suited to the city's purchasing power. Their stalls are long-established, and many vendors trade alongside relatives. This low-churn setting suits a study of the expectations and social ties that shape relocation decisions, without the confounding variation a transient vendor population would introduce.

Third, the SVA's implementation failures are particularly visible in Patna. Bihar is among India's most populous and least developed states, and Patna's street vending sector is large: PMC records count about 30,000 vendors. PMC has constructed several vending zones under the SVA, but implementation has stalled. All but one record zero occupancy, and the exception is only a quarter occupied. Figure 1 shows their locations. The zone designs also fix the space allocated to each vendor. Under the Bihar implementation of the Act, each spot measures either 6 by 4 feet or 5 by 3 feet. We return to this in Section 4.6.

Figure 1: Proposed and Constructed Vending Zones across the City



Note: Officially recorded street vending zones in Patna, by construction stage. Thirty-three appear in a government record: a SULM portal entry, the 2021 Urban Development & Housing Department sanction (Memo 11913/2021), or a record of construction. Thirty are plotted, of which 11 are built, 2 are under construction and 10 are in planning; a further 7 are reported built on portal entries whose status field and status text disagree. The remaining three could not be located. Coordinates come from the SULM portal (10) and geocoded landmarks (20). One zone is designated as a stretch of road rather than a point and is drawn along the road.

2.2 Primary Survey

We collected the data in three stages: marketplace listing, vendor recruitment and the main survey. Patna is divided into 75 wards. We selected the 40 wards with the highest concentration of vendors, using the 2021 National Urban Livelihoods Mission (NULM) survey of vendors obtained from PMC. To rank wards, we geocoded potential vending locations and constructed vending zones, and overlaid them on ward boundaries.

Marketplace Listing. In August 2024, we listed all marketplaces in the 40 study wards. These wards contain about 17,000 stationary vendors, the substantial majority of Patna's stationary vendor population. For each marketplace, we recorded its geolocation, the characteristics of the surrounding area and an approximate count of vendors. We define a marketplace as any public space where more than 10 vendors cluster to sell goods or services. We define a street vendor as a person, typically self-employed, who sells goods or services from a temporary structure at a fixed location in a public space, such as a street, lane, sidewalk, footpath, pavement or public park. We exclude hawkers and mobile vendors. This definition is narrower than the ILO definition in Section 1, which includes mobile vendors. The experiment requires the restriction: the choice tasks price attributes of a fixed location, and rent, number of spots and commuting distance have no counterpart for a vendor who does not occupy one. We target produce and street food vendors, who together make up 66 per cent of the street vending sector according to EBRD informal enterprise surveys. The listing identified 375 marketplaces across the 40 wards, which we reduced to 206 analytical marketplaces in two steps. First, before the survey, we excluded 43 marketplaces: those whose primary product was not produce or street food, and one that the city had declared a commercial zone allocated on a first-come-first-served basis. This left 332 shortlisted marketplaces. Second, we merged 126 of these into nearby markets to protect the information experiment. Because we assign treatment within marketplaces, contamination depends on the time between surveying

a treated vendor and surveying a nearby control vendor. Merging adjacent clusters into one marketplace meant we surveyed each within a single day, leaving hours, not days, for information to spread. Had we kept them as separate units, we would have visited them on different days, and vendors in the later cluster could have heard about the Act from those surveyed earlier.

Recruitment and Survey Administration. We returned to the 206 marketplaces between June and October 2025. The survey proceeded in two stages. First, we made recruitment visits during peak marketplace hours to identify eligible vendors and schedule interviews at times when they were relatively free. Enumerators followed a systematic random walk, approaching every third vendor in each marketplace who sold produce or street food from a fixed location. We recruited 3,499 vendors.

Second, within a few days of recruitment, enumerators administered the main survey in person. It typically lasted 45 minutes. Of the 3,499 recruited vendors, 2,935 completed the main survey, a response rate of 84 per cent. We exclude 13 incomplete responses, which leaves an analytical sample of 2,922 vendors across 206 marketplaces. All descriptive statistics in this section use this sample. The discrete-choice estimation in Section 4 uses 2,910 of these vendors across 203 marketplaces: we drop 12 vendors with a missing covariate, and with them three marketplaces (Section 3.2). No descriptive statistic reported here changes by more than a tenth of a percentage point between the two samples. Respondents are similar to non-respondents in age, trading location and stall type. They are more likely to sell produce (59.2 against 49.5 per cent) and slightly less likely to be male (88.3 against 91.0 per cent) (Appendix A3).

Figure 2, panel (a), maps the locations of all 2,922 vendors. Vendors cluster strongly along transport corridors. Panel (b) maps vendors' reported residences, geocoded from the landmarks they named to the centroid of the surrounding locality. Residences are dense in the central wards but extend much further east, so a sizeable share of vendors commute inward to their stalls. Panel (c) maps the markets that vendors name as their source of supply. Wholesale (produce) markets sit mostly on the periphery of the built-up area, while ingredient (retail) markets sit in the centre, beside the densest vendor clusters. This pattern suggests that bulk produce enters the city at its edges and is redistributed inward. Most vendors live close to their place of business: 89 per cent live within 3 km (Figure 3, panel (a)). And 91 per cent pay no fee of any kind for their current spot (Figure 3, panel (b)).

Table 1 presents descriptive statistics for the analytical sample. Vendors are on average 39 years old with 6 years of education. Eighty-eight per cent are male, and 96 per cent belong to historically disadvantaged caste groups: 87 per cent to Other Backward Classes and 10 per cent to Scheduled Castes or Tribes. For most vendors, street vending is not transitional work. Fifty-three per cent have vended for over 10 years, and 95 per cent work 7 days a week, for 9 hours a day on average. Average daily profit is INR 619, and average daily sales are INR 2,991. Eighty-five per cent operate from stalls or carts. Ninety per cent trade along roads or streets and the remaining 10 per cent on pedestrian walkways, which leaves them highly exposed to eviction.

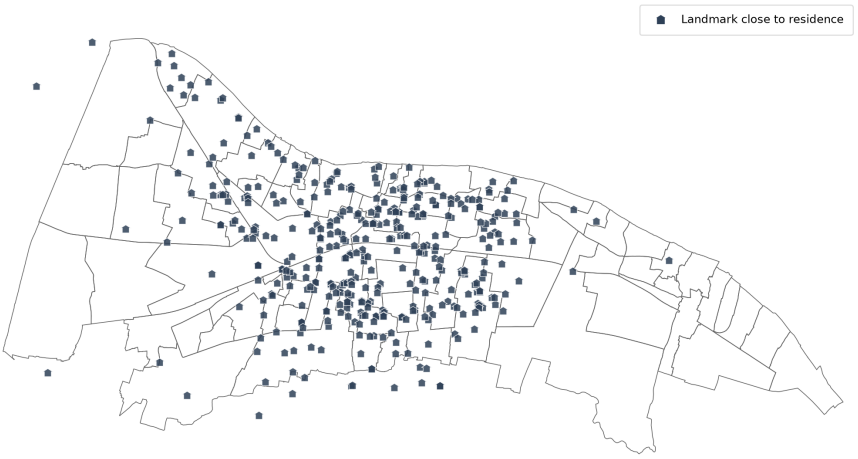
Figure 4 describes the licensing and enforcement environment. Panel (a) shows that

Figure 2: Geo-location of vendors, their homes and sourcing markets

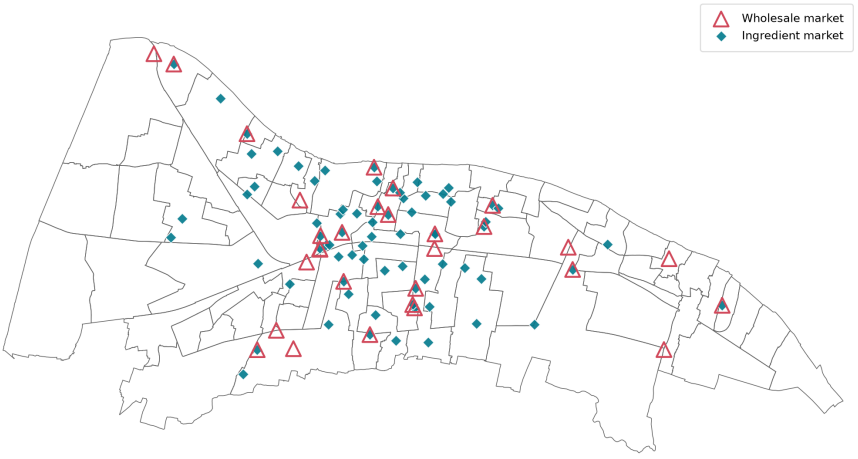
(a) Vendors' Business



(b) Vendors' Residence



(c) Wholesale and Inventory Markets



Note: (a) Surveyed vendors across the 40 study wards. (b) Each marker is a distinct landmark reported as near a vendor's home, geocoded to its locality centroid; 412 of the 719 landmarks reported were geocoded confidently and fall within the mapped area. Residence locations are available for 2,374 of 2,922 vendors (81.2 per cent), plus 118 who live near their stall and take its coordinates; the remainder gave no answer or a landmark that could not be tied to a place name. (c) Markers are distinct markets named by vendors: outlined triangles are wholesale (produce) markets, filled diamonds ingredient markets. Fourteen markets outside the mapped area are included in all counts but not drawn. Market locations are available for 1,805 vendors (61.8 per cent), covering 84.0 per cent of fruit and 79.5 per cent of vegetable vendors but only 32.9 per cent of cooked-food vendors, who mostly source from neighbouring shops or take deliveries.

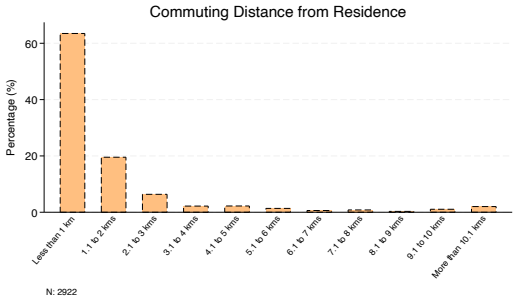
Table 1: Summary Statistics

Variable	Mean	SD
Age	38.64	12.76
Education	6.08	4.76
Male	0.88	0.32
Marital Status		
Married	0.82	0.38
Single	0.16	0.36
Widowed/Divorced	0.02	0.15
Caste Group		
Backward Caste	0.86	0.34
Scheduled Caste/Tribe	0.10	0.30
General	0.04	0.19
Product Category		
Fresh Produce	0.59	0.49
Cooked Food/Drinks	0.41	0.49
Premises Type		
On Main Roads or streets	0.90	0.30
On Pedestrian Walkways	0.10	0.30
Infrastructure Type		
Have Stalls/Carts	0.85	0.36
On the Floor	0.15	0.36
Other Business related		
Work 7 days a week	0.95	0.22
Hours of Operation	9.22	3.67
Have employees	0.32	0.46
Yesterday's Profits (INR)	618.53	626.31
Weekly Profits (INR)	4,259.65	4,056.31
Monthly Profits (INR)	18,458.50	17,577.33
Own Other Businesses	0.15	0.35
Never Changed Marketplaces	0.99	0.08
Years in Business		
Upto 1 year	0.06	0.24
1 to 5 years	0.21	0.41
5 to 10 years	0.20	0.40
More than 10 years	0.53	0.50

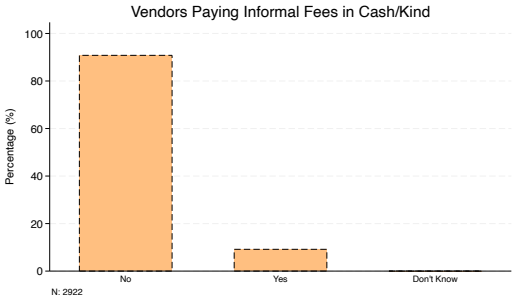
Note: N = 2922. All monetary units are expressed in Indian Rupees (INR). Monthly profits are constructed from reported weekly profits as weekly profits $\times$ 52/12. Profits were elicited for the previous day and the previous week only, since most vendors do not keep written accounts and longer recall windows are unreliable.

Figure 3: Status-quo Environment

(a) Reported Commuting Distance between Residence and Marketplace



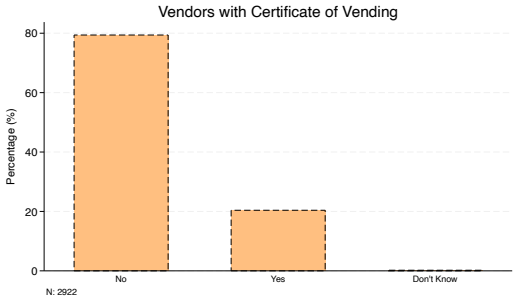
(b) Share of Vendors Paying Informal Fees at Current Business Location



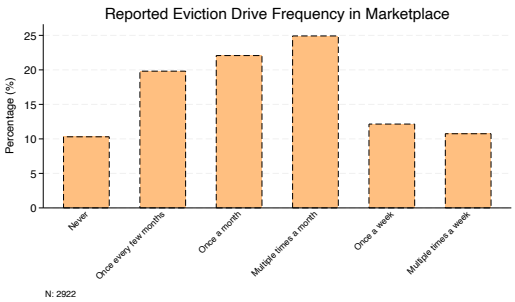
20 per cent of vendors hold a certificate of vending, so certification under the SVA has reached a fifth of the sample. Panel (b) reports how often eviction drives (*halla gadis*) occur in the vendor's marketplace. Ten per cent report that they never occur, 70 per cent report at least one a month, and 23 per cent at least one a week. Panels (c) and (d) record personal exposure over the six months before the survey: 42 per cent report harassment by officials or the police, and 33 per cent report an eviction or confiscation of goods. Enforcement is routine, and it reaches vendors directly, not through fines or paperwork.

Figure 4: Licensing and Enforcement Environment

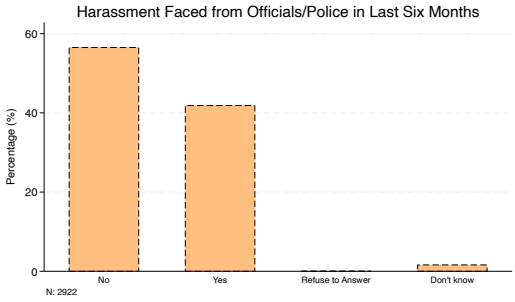
(a) Certificate of Vending



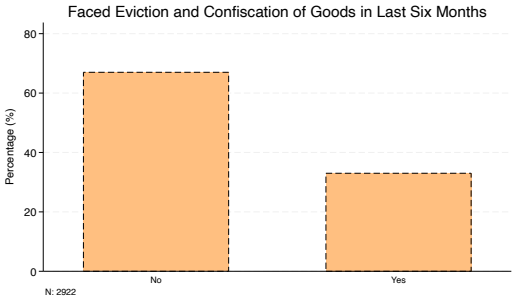
(b) Eviction Drives Frequency in Marketplace



(c) Harassment from Officials or Police



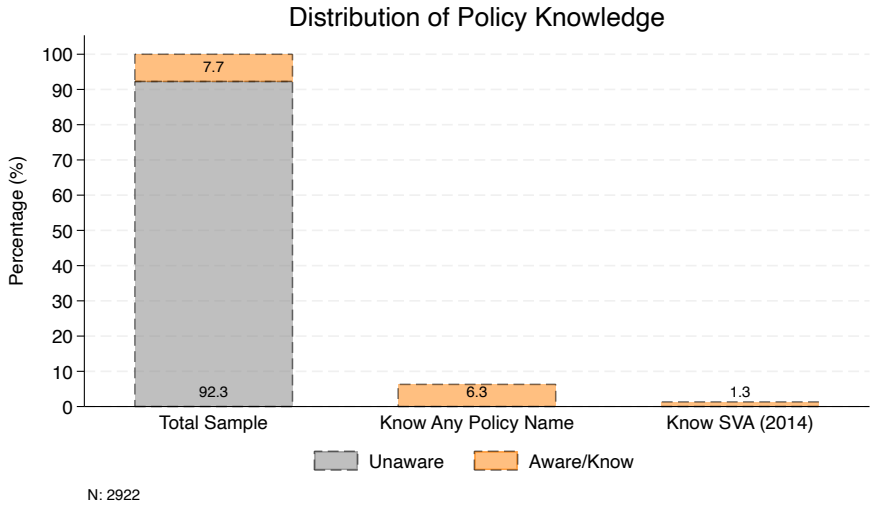
(d) Eviction or Confiscation of Goods



Yet awareness of the policy that would protect vendors is close to zero. Only 1.3 per cent could identify the Street Vendors Act (Figure 5). Only 7.7 per cent were aware of any policy on street vending, and 6.3 per cent could name one. Active enforcement combined

with near-zero policy awareness motivates both the study setting and the information experiment in Section 3.3.

Figure 5: Awareness of Policies Among Street Vendors



3 Empirical Strategy

To price the relocation the Act requires, we need to recover how vendors value the terms on which formality is offered: the rent charged, the distance from their residence, the number of spots, the competition and demand they expect to find, and the share of peers who also move. Three features of the setting prevent us from recovering these valuations from observed behaviour. First, 99 per cent of vendors have never changed marketplace, so revealed-preference data contain almost no variation in location choice. Second, the terms do not vary in the field. PMC has constructed few zones, and they are almost entirely unoccupied (Section 2.1), so there is neither a spread of offers to compare nor take-up against which to price them. Third, the peer share is simultaneously determined: each vendor's relocation decision depends on the share s of peers relocating, while s is itself the aggregate of those decisions. Observational estimates would also confound the causal effect of peer behaviour with common unobservables that drive both.

We address all three problems by assigning the terms of the offer instead of observing them. A stated-preference experiment presents each vendor with a sequence of hypothetical zones whose attributes are drawn from a designed distribution, so rent, distance and the peer share vary independently of vendor characteristics. The approach is well established in transportation and environmental economics, and Mas and Pallais (2017) apply it to labour market choice. Two features of our implementation limit hypothetical bias. First, we personalise the scenarios: every attribute is expressed relative to each vendor's verified current conditions, which anchors choices to circumstances the respondent already knows (Taylor and Cummings 1999). Second, we told vendors at the outset that their responses would inform PMC policy. Appendix A6.2 describes these and our other responses to hypothetical bias, inattention and fatigue. A randomised information treatment, described in Section 3.3, is embedded between choice rounds.

3.1 Stated-Preference Experiment

Cognitive Testing. We tested the vignette with about 50 street vendors in Patna before finalising it. Many street vendors have low literacy, so the format had to work for every respondent regardless of reading ability. We presented draft choice tasks and asked for feedback on the framing of the question, the language describing attribute levels, and the visual presentation of the two alternatives. We then refined the design over several iterations, adding visual aids to represent attributes and to help respondents compare the status quo with the alternative. Appendix Figure A2 presents the final format.

Experiment Design. Each respondent completed six choice tasks. The first was a practice round (round 0) to familiarise participants with the format, and we exclude it from the analysis. Rounds 1–5 form the analytical sample. In each task, the respondent chose between two alternatives: remaining at the current location or relocating to a vending zone. The same six attributes describe both alternatives. The zone's attribute values vary systematically across choice occasions, but the zone always offers the legal right to operate, the feature that distinguishes formal from informal vending.

Enumerators told vendors that the new marketplace was identical to their current one except in the attributes shown, and that storage capacity and transport costs were unchanged. This holds fixed the supply-side conditions that would otherwise confound valuations of the offered terms, so differences across profiles operate only through the attributes we vary. It also holds each vendor's scale of operation fixed by construction, a limitation we return to in Section 4.6. Appendix A6.3 reproduces the enumerator script.

Attribute Selection and Levels. We selected the six attributes from formative qualitative interviews and the literature on informal-sector formalisation. Table 2 lists the attributes and their possible values in the vending zone. We set attribute levels through pilot testing, so that they reflect realistic variation in actual vending markets while spanning enough range to identify preferences. Appendix A6 explains how we selected the scenarios.

Table 2: Possible Values of Attributes in the New Marketplace

Attributes	Levels
Monthly Rent (% of monthly sales)	0, 1, 2, 4, 8
Number of spots	same, 20, 40, 80, 160
Expected number of vendors relocating (% of vendors)	[0, 100]
Expected number of competitors (absolute change)	-10, -5, 0, +5, +10
Expected number of daily customers (% change)	-50, -20, 0, 20, 50 if current $\in$ [10-60] -30, -15, 0, 15, 30 if current $\in$ [61-120] -25, -10, 0, 10, 25 if current $\in$ [121-170]
Distance from residence (absolute change)	-1, 0, +1, +3, +5

The zone's attribute values are defined *relative to each vendor's self-reported status quo*, elicited in the survey module before the choice tasks. For example, if a vendor reported

a 2 km commute and the design specified +1 km, the card showed a 3 km commute. All vendors therefore face the same experimental variation in relative terms, while the absolute values reflect their actual situation. All pairwise attribute correlations in the design matrix are below 0.2, so we can estimate individual attribute effects without multicollinearity concerns (Appendix A6).

Two features of the assignment limit what the design can identify. First, we floored reductions in commuting distance at each vendor's existing commute, so an offer of a commute one kilometre shorter often delivered no change. The experiment therefore identifies the cost of a longer commute but not the value of a shorter one. Second, we constructed the assigned peer share by shifting each vendor's own elicited belief by a randomised shock of −50, −20, 0, +20 or +50 percentage points. We rounded the result to the nearest ten points, because the card presents the share as a count out of ten vendors, and censored it at zero and 100. The exogenous variation actually delivered is therefore narrower than the design's nominal range. Section 4.5.2 reports both.

3.2 Discrete Choice Model

In round t, vendor i chooses between remaining at the current site and relocating to the offered zone, and relocates if the net utility gain is positive:

$$U_{it}^{\text{relocate}} - U_{it}^{\text{stay}} = V_{it} + \varepsilon_{it} > 0, \tag{1}$$

with ε_{it} logistic and independent across i and t. The deterministic component is

$$V_{it} = \bar{\alpha} + z_i'\gamma + a_i + x_{it}'\beta + d_i \cdot \text{km}_{it}, \tag{2}$$

so the model is a binary panel logit with random coefficients. The vector x_{it} collects the attributes of the offer, which vary across occasions by design, together with round dummies and the information treatment interacted with post-treatment rounds. The vector z_i collects 27 vendor characteristics measured before the choice tasks, and each vendor's elicited belief about the share of peers who would relocate.

Functional form is tested, not imposed. Every attribute takes a small number of discrete levels, so each can enter as a full set of level dummies. This saturated specification imposes no shape on any attribute and assumes only additivity across them. Every simpler form is a restriction of the saturated model, and we test each with a nested likelihood-ratio test. Commuting distance enters as free dummies at the offered levels,

$$\sum_{\ell \in \{-1,\,+1,\,+3,\,+5\}} \beta_\ell^D \cdot \mathbf{1}\{\Delta\text{Distance}_{it} = \ell\}.$$

The data reject a linear penalty in distance ($p = 0.041$), and the free dummies nest every shape the offered levels can identify. Rent, assigned as a percentage r_{it} of each vendor's own monthly sales, enters as a linear spline with knots at one and two per cent,

$$\beta_1^R \min(r_{it}, 1) \;+\; \beta_2^R \big(\min(r_{it}, 2) - 1\big)^+ \;+\; \beta_3^R (r_{it} - 2)^+.$$

The spline is the free-dummy model subject to a single restriction, which the data do not reject. The remaining attributes (the number of spots, expected competitors, expected

customers and the assigned peer share s) enter linearly; we test each against its own free dummies. Appendix A7 reports every test, including the test of a fully linear specification, which the saturated model rejects. We maintain additivity across attributes and do not test it. The design was not constructed to identify interactions between distance and rent, so the quantities in (5) rest on additivity as an assumption.

Vendors differ. Two coefficients are random, each estimated as an independent normal. The vendor-specific intercept $a_i \sim \mathcal{N}(0, \sigma_a^2)$ is the private value of a zone: the part that observed characteristics do not explain. The vendor-specific slope $d_i \sim \mathcal{N}(0, \sigma_d^2)$ lets each additional kilometre of commute cost vendors different amounts. The mean of d_i is zero by construction, not by estimation. No continuous distance term enters the fixed part of (2), so the free dummies β_ℓ^D carry the entire average response to distance, and σ_d measures dispersion around a mean response whose shape is unrestricted. Section 4.1.2 shows why the model needs a random intercept, and Appendix A8 shows why it needs a random slope. The appendix also reports the alternatives, including a latent class model and non-parametric distributions for a_i.

The likelihood integrates (2) over a_i and d_i with a product rule: a 241-node rectangle rule spanning six standard deviations for the intercept, crossed with twelve-point Gauss–Hermite quadrature for the distance slope. We do not use Gauss–Hermite for the intercept, because its dispersion is large relative to the logistic scale ($\hat{\sigma}_a \approx 4.9$) and a Hermite rule places too few nodes where the mass lies. On a much finer grid, each reported log-likelihood moves by less than 0.01. We start every model from three values of each standard deviation, and all three reach the same optimum. We cluster standard errors at the marketplace, the level at which we assigned the choice scenarios. We also report a 200-replication marketplace pairs bootstrap. The two methods imply the same inferences for every quantity in Section 4, and their interval widths differ by at most a third. The estimation sample covers all five rounds for 2,910 vendors: 14,550 choice occasions across 203 marketplaces. The 12 vendors dropped from the surveyed sample each lack a covariate in z_i, and dropping them removes three marketplaces.

Quantities. The choice probability for a vendor with given (a_i, d_i) is

$$P(\text{relocate}) = \frac{\exp(V_{it})}{1 + \exp(V_{it})}. \tag{3}$$

Predicted take-up for any zone design averages this probability over vendors and over the distribution of the random coefficients. For attributes other than rent, we report trade-offs in kilometres of commute, using the per-kilometre rate between the +1 and +3 km levels:

$$\text{Trade-off}_k = \frac{\hat{\beta}_k}{\left(\hat{\beta}^D_{+3} - \hat{\beta}^D_{+1}\right)/2}. \tag{4}$$

Each trade-off is a ratio of coefficients within one model, so it does not depend on the scale normalisation. We report these ratios alongside the same ratios from a fixed-effects conditional logit, which assumes nothing about how vendors differ.

Rent is not linear below two per cent of sales, so we cannot state the cost of a move by dividing a distance coefficient by a rent slope. Instead, we report the maximum rent a

zone can charge at each offered distance. Let $W_i(\ell) = \bar{\alpha} + z_i'\gamma + a_i + \beta_\ell^D + \tilde{x}_{it}'\tilde{\beta}$ denote utility at distance ℓ, with every attribute other than rent at the vendor's status quo. The maximum acceptable rent $r_i^*(\ell)$ solves

$$\beta_1^R \min(r, 1) + \beta_2^R \big(\min(r, 2) - 1\big)^+ + \beta_3^R (r - 2)^+ = -W_i(\ell), \tag{5}$$

which has a closed form within each segment of the spline. We report r^* for the average vendor and as a distribution across vendors. Two groups have no r^* the experiment can speak to: vendors who refuse the zone even at a rent of zero, for whom (5) has no solution, and vendors whose solution lies above eight per cent of sales, the highest rent the design offered. Quartiles are therefore taken over the vendors whose maximum falls inside the offered range, with the size of both excluded groups stated beside them. The share of vendors who accept at each offered rent, in Section 4.5.1, needs no such restriction.
Because the elicited peer belief enters z_i, the intercept alone is not the value of a zone. We evaluate every valuation at the sample-mean belief and also report the value at a belief of zero. Section 4.2 explains why we retain the belief despite this. Without it, the assigned peer share inherits the construction from which it was generated, and the shock no longer identifies the coefficient we use to bound coordination effects.

3.3 Information Experiment

The awareness data in Section 2 motivate the information experiment. At the start of the survey, we randomly assigned vendors within each marketplace to treatment and control groups. After round 3 of the choice tasks, enumerators gave treated vendors detailed information about the Street Vendors Act (2014) and concrete evidence of its enforcement in Chandigarh, where about 3,000 non-compliant vendors were displaced in late 2024. Appendix A6.4 reproduces the full treatment text. Enumerators delivered the information orally to accommodate low literacy. Control vendors received no information and proceeded directly to round 4. All vendors then completed two further choice tasks (rounds 4–5) in the same format. Before treatment, the two groups are balanced on observable characteristics, except for a difference in weekly sales significant at the 10 per cent level (Appendix Table A5).

We allow the treatment to shift only the intercept of (2), and constrain the attribute slopes to be common across treatment and control:

$$V_{it} = \bar{\alpha} + \underbrace{\lambda \cdot \text{treat}_i \cdot \text{post}_t}_{\text{treatment effect}} + \sum_{t=2}^{5} \rho_t \mathbf{1}\{\text{round} = t\} + z_i'\gamma + a_i + x_{it}'\beta + d_i \cdot \text{km}_{it}, \tag{6}$$

where $\text{post}_t \in \{0, 1\}$ indicates rounds 4–5 and $\text{treat}_i \in \{0, 1\}$ indicates treatment assignment. The coefficient of interest is λ. We do not estimate a separate main effect of treat_i: assignment was randomised within marketplace, so $z_i'\gamma$ and the random intercept a_i absorb any time-invariant difference between the arms. The round dummies ρ_t absorb the main effect of post_t. The treatment tells vendors that the policy will be enforced and that informal sites will be cleared. It should therefore raise the value of relocating relative to staying, without changing the relative weights vendors place on the zone's design attributes. We test this restriction instead of assuming it. In a conditional-logit specification of the same design, interacting the treatment with every design attribute does not reject the restriction that the treatment shifts the intercept alone ($\chi^2(8) = 5.82$,

$p = 0.67$). A further test, which allows the treatment effect to differ across the range of estimated valuations, does not reject a common effect ($p = 0.18$).

Because the treatment shifts only the intercept, we pool all five rounds in a single estimation, and (6) is the specification we report throughout Section 4. The pre-treatment rounds identify the trade-offs between design attributes, and the post-treatment rounds identify λ. The round dummies absorb learning, fatigue and any drift common to both arms. We convert the treatment effect to percentage points of monthly sales through (5) and to kilometres through (4).

4 Results

This section establishes three main findings. First, vendors value formality: at an unchanged commute, the average vendor's willingness to pay for a protected spot is 4.5 per cent of monthly sales, and predicted take-up of a free spot is 68 per cent. Second, distance consumes that value. Willingness to pay falls to 2.1 per cent of sales after a commute one kilometre longer and reaches zero at about 3.8 kilometres, and a five-kilometre move lowers take-up twice as much as a rent of two per cent of sales. Third, information about the Act raises the value of relocating without changing how vendors trade off the zone's attributes: it lifts predicted take-up by about 3 percentage points at every design, but it cannot offset the cost of distance. Section 4.1 first sets out the estimation choices behind these results. Sections 4.2 to 4.4 establish the three main findings. Section 4.5 reports secondary results on rent, coordination among vendors, zone size and heterogeneity across vendors, and Section 4.6 discusses the one margin the design cannot price.

We measure willingness to pay in percentage points of monthly sales, because the numéraire is the rent slope above two per cent of sales. The design assigned rent as a percentage of each vendor's own reported sales, so the recovered parameters are rates on own sales, not rupee amounts for a representative vendor. We convert to rupees at median monthly sales in the sample, INR 60,667, so one percentage point of sales is INR 607. Intervals are from the 200-replication marketplace pairs bootstrap (Section 3.2).

Two distinct shares appear below. Predicted take-up averages each vendor's choice probability over the estimated distribution of the random coefficients. Table 7 reports it, and we use it whenever we quote a share of vendors who would relocate. The share of vendors with a positive valuation counts the vendors for whom the zone is worth more than the status quo in the deterministic part of utility, ignoring the logistic error. Figure 14 traces this share across rent levels. The logistic error pulls predicted take-up towards one half, so the share with a positive valuation is the larger of the two whenever it exceeds one half. We name the measure at each use.

4.1 Estimation

Three estimation choices underlie every result below: the functional form of distance and rent, the treatment of unobserved heterogeneity, and the main specification.

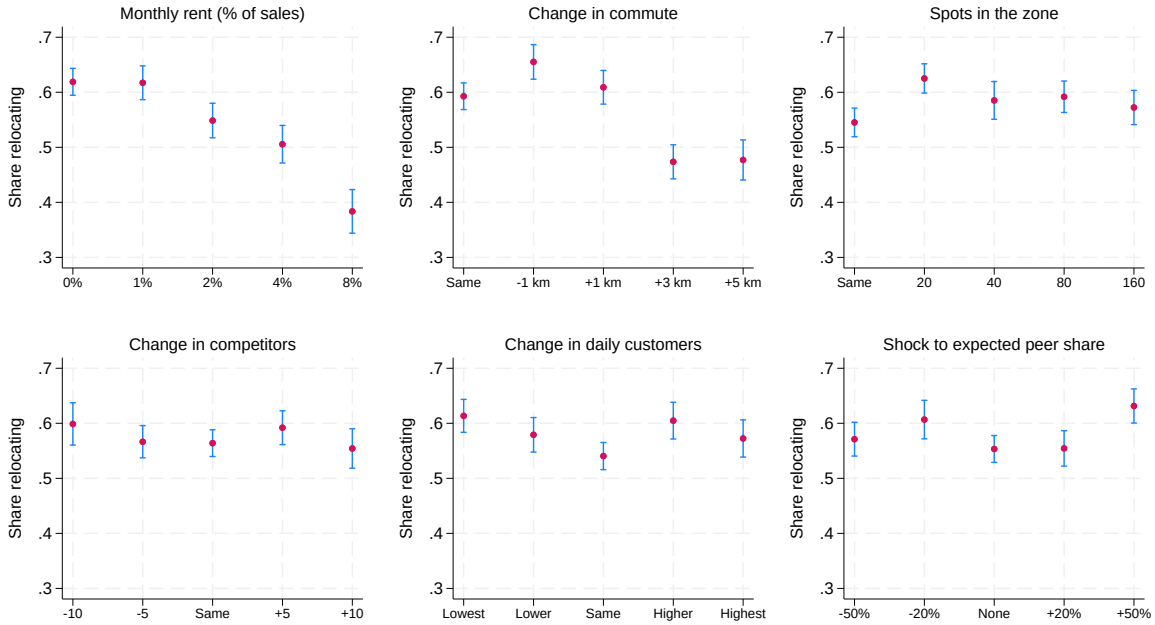
4.1.1 Functional Form

Every attribute takes a small number of discrete levels, assigned at random, so we can read the relocation rate at each level without a model. Figure 6 plots these raw shares for all six attributes, with marketplace-clustered confidence intervals. Figure 7 reports the same levels in a fixed-effects logit with a free dummy at each offered level, which imposes no shape on any attribute and no distribution on vendor intercepts.

Likelihood-ratio tests against the saturated model reject linearity in both distance and rent (Appendix A7). For distance, the data reject a linear penalty ($p = 0.041$). The penalty for a commute one kilometre longer is 0.43 of the penalty for a commute three kilometres longer (95% CI [0.31, 0.56]). Linearity implies one third, which lies inside this interval, so the step ratio alone cannot distinguish the two. For rent, the data reject both a rent that enters linearly from zero ($p = 0.008$) and a rent that is free below one per cent of sales ($p = 0.039$).

We therefore carry both attributes into the estimation in the least restrictive form the design supports. Distance enters as free dummies at each offered level. Rent enters as a linear spline with knots at one and two per cent of sales: the free-dummy model subject to a single restriction, which the data do not come close to rejecting ($p = 0.95$). The rent ladder reported below assumes that distance and rent enter additively (Section 3.2).

Figure 6: Relocation Rate at Each Attribute Level

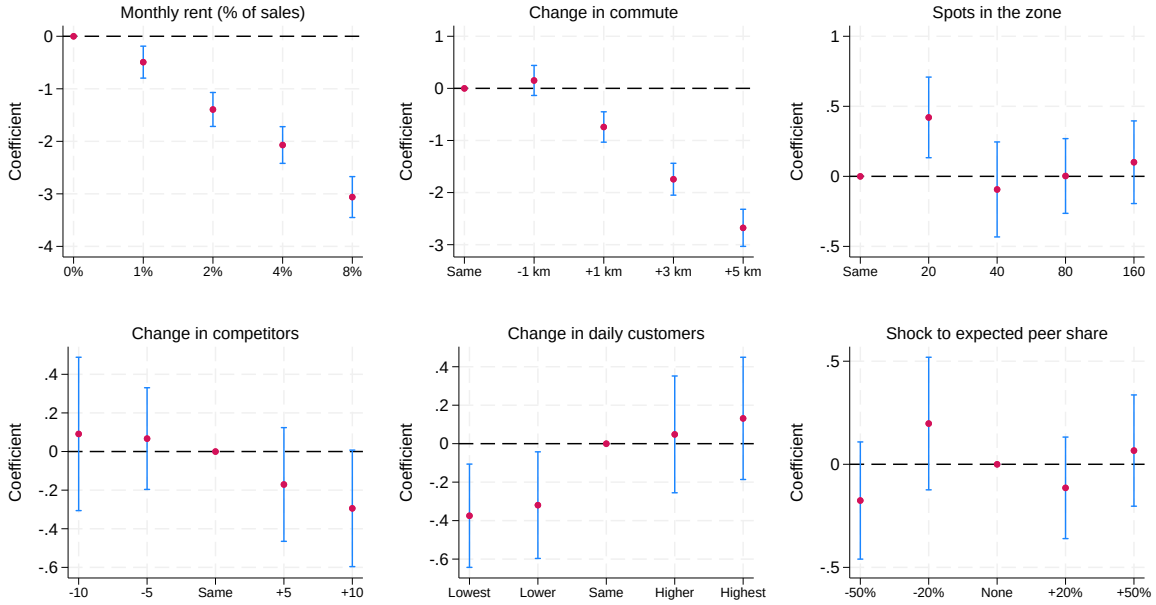


Raw shares over all five rounds; 95% CIs clustered by marketplace.

4.1.2 Unobserved Heterogeneity

A model in which all vendors share the same preferences cannot reproduce the data. Figure 8 plots the observed distribution of the number of rounds, from zero to five, in which a vendor relocates, against the distribution a homogeneous conditional logit predicts. The

Figure 7: Attribute Levels in a Fixed-Effects Logit with Free Dummies



FE logit with vendor fixed effects and free level dummies;
95% CIs clustered by marketplace. Base: no change.

data are concentrated at the two ends: 748 vendors never relocate and 1,021 always do. The homogeneous model places far too many vendors in the middle.

Inattention does not explain the vendors whose choices never vary. Only 4.2 per cent of respondents failed the embedded attention check, and failure rates do not differ across the range of estimated valuations ($p = 0.13$). Vendors who refuse the zone in every round score no lower on cognitive ability than those who accept in every round. If yea-saying drove the unconditional acceptances, the failure rate would stand out among those vendors, and it does not.

Allowing for unobserved heterogeneity repairs the fit and is worth far more than the covariates. Table 3 reports held-out fit on the control group's final two rounds, with each vendor's posterior formed from their training rounds alone. With the 27 vendor characteristics already in the model, allowing vendors to differ improves the held-out log-likelihood by a further 760 points, against about 29 from the characteristics themselves.

Table 4 reports two variance figures that measure different things. The 27 characteristics generate a predicted spread in the value of a zone with a variance of 4.43, which is 15.5 per cent of the total spread across vendors. Yet adding them reduces the *unexplained* variance by only 6.1 per cent. Together, these figures say that the characteristics shift where a vendor sits on average, but explain little of why a vendor differs from others with the same characteristics. Section 4.5.4 returns to what this implies for targeting.

We therefore estimate a mixed logit with two independent random coefficients, a vendor-specific valuation of the zone and a vendor-specific per-kilometre cost (Section 3.2). Appendix A8 reports the alternatives we estimated, including a three-class latent class model and flexible non-parametric distributions of valuations. None of them improves fit enough

to justify its additional parameters.[1]

Figure 8: Observed and Predicted Distribution of Relocation Counts

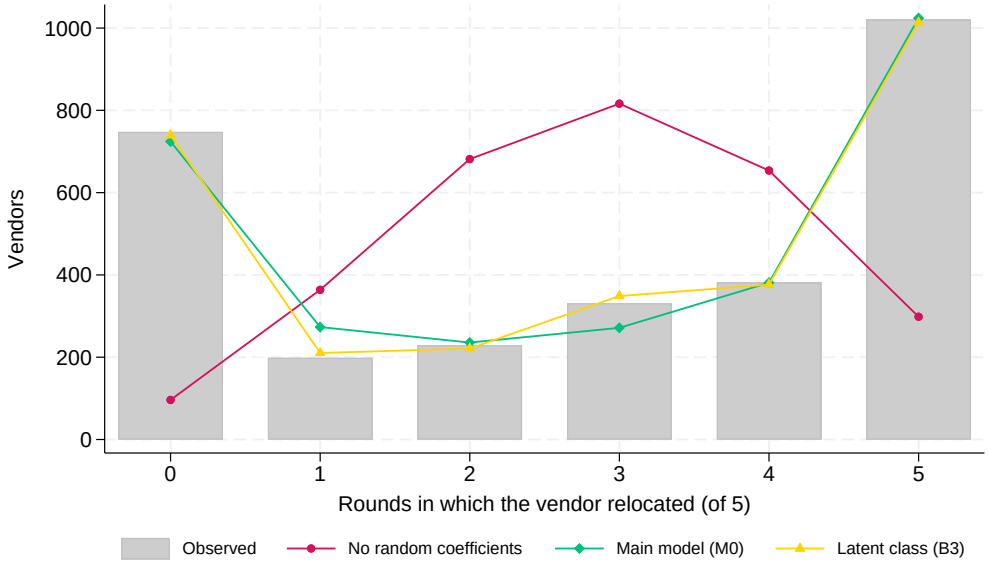


Table 3: Held-out Fit

	Log-likelihood	Brier	Hit rate
Conditional logit	-1827.3	0.224	0.638
Conditional logit + 27 covariates	-1798.8	0.219	0.647
Mixed logit + 27 covariates	-1037.9	0.110	0.854
Latent class, three types	-1063.0	0.114	0.847
Gain in held-out log-likelihood			
The 27 covariates alone	28.5		
Heterogeneity, on top of the covariates	760.8		
Both together	789.4		

Notes: Fit on the control group's final two rounds, which no model saw in estimation; each vendor's posterior is formed from their training rounds alone. Brier is the mean squared prediction error and the hit rate the share of choices called correctly, both on the same held-out rounds. The lower panel decomposes the gain over the bare conditional logit: the covariates are worth little, and almost all of the improvement comes from allowing vendors to differ once the covariates are already in.

[1]One richer specification does. Adding a vendor-specific cost of changing site at all improves the likelihood by 11.3 log points for one parameter, and it is the only specification that also improves fit out of sample. It changes the maximum rent at an unchanged commute, the break-even distance and the value of information by less than one per cent, and moves the intermediate rungs of the ladder by at most eight per cent. It differs materially in one respect: the specification used here understates the dispersion of the distance response over short moves by a factor of about 2.5. Statements below about how widely vendors differ at short distances are therefore lower bounds. Appendix A8 reports the model in full.

Table 4: What the Observed Characteristics Explain

	Value
Fitted SD of the random intercept	4.91
SD of the covariate-predicted intercept	2.11
SD of the posterior intercept	4.20
Share of zone-value variation from covariates (%)	15.5
Fall in unexplained variance when covariates added (%)	6.1

Notes: All quantities are in the utility scale of the zone intercept. The two percentages measure different things and are reported together for that reason: the covariates generate a predicted spread that is 15.5 per cent of the total variation across vendors, yet adding them reduces the *unexplained* variance by only 6.1 per cent. They shift where a vendor sits on average without explaining why that vendor differs from others like them. The 15.5 per cent is measured against the total variation in the value of a zone, which is the sum of the two variances above it, $4.91^2 + 2.11^2 = 28.6$.

4.1.3 Main Estimates

Table 5 reports four specifications. Column (1) is a conditional logit in which rent and distance enter linearly. Column (2) replaces rent and distance with free level dummies, and the likelihood-ratio test of (1) against (2) rejects linearity ($\chi^2(6) = 19.94$, $p = 0.003$). Column (3) is a fixed-effects conditional logit in the adopted forms. It imposes no assumption on how vendors differ, at the cost of using only the vendors whose choices vary across rounds. Column (4) is the main specification, and every quantity below comes from it. Coefficient levels in column (3) are on a different scale from column (4), because the two models treat intercept heterogeneity differently. Ratios of coefficients do not depend on scale, and the trade-offs in kilometres of commute agree. Read off column (3), a percentage point of rent is worth 1.09, 1.68 and 0.58 kilometres of commute in the three segments of the spline, and the information treatment is worth 0.77 kilometres, against 0.99, 1.84, 0.64 and 0.86 from column (4). None of the four differences is significant.[2] Appendix Table A4 reports the covariate coefficients.

[2]Column (3) uses the 1,141 vendors whose choices vary across rounds, the only vendors that identify a fixed-effects logit; the other columns use all 2,910. Its log-likelihood is conditional, so it has no BIC.

Table 5: Main Estimates

	(1) Rent/dist linear	(2) Rent/dist free	(3) FE logit	(4) Main model (M0)
main				
Rent (% of sales)	-0.175***			
	(0.013)			
Rent, 0-1% (per point)			-0.550***	-0.581***
			(0.132)	(0.146)
Rent, 1-2% (per point)			-0.849***	-1.082***
			(0.128)	(0.146)
Rent, above 2% (per point)			-0.296***	-0.376***
			(0.028)	(0.034)
1%		-0.215***		
		(0.067)		
2%		-0.551***		
		(0.073)		
4%		-0.833***		
		(0.090)		
8%		-1.428***		
		(0.108)		
Change in commute (km)	-0.189***			
	(0.016)			
-1 km		0.053	0.277	0.506**
		(0.120)	(0.213)	(0.234)
+1 km		-0.326***	-0.783***	-0.900***
		(0.068)	(0.129)	(0.147)
+3 km		-0.773***	-1.796***	-2.074***
		(0.069)	(0.138)	(0.175)
+5 km		-0.993***	-2.735***	-3.393***
		(0.089)	(0.166)	(0.262)
Information arm	-0.009	-0.005		
	(0.069)	(0.069)		
Information x post	0.184***	0.185***	0.392**	0.506***
	(0.055)	(0.055)	(0.152)	(0.175)
SD: intercept				4.912***
				(0.201)
SD: per-km slope				0.725***
				(0.049)
Log likelihood	-8968.88	-8958.91	-1614.13	-6005.87
Parameters	59	65	16	53
BIC (on vendors)	18408.34	18436.26		12434.47
Vendors	2910	2910	1141	2910
Choice occasions	14550	14550	5705	14550
Marketplaces	203	203	182	203
LR: rent and distance linear vs free		19.94		
LR degrees of freedom		6		
LR p-value		0.0028		

Notes: Standard errors clustered at the marketplace in parentheses. Column (1) enters rent and distance linearly; column

4.2 Vendors Value Formality

The first main finding is that vendors place a substantial value on formality itself. Suppose a municipality designates an existing site as a vending zone, so that no vendor's commute changes. The average vendor's willingness to pay for a spot is then 4.48 per cent of monthly sales (95% CI [3.34, 5.68]), or INR 2,718 per month at median sales. At a free spot on these terms, predicted take-up is 67.9 per cent ([65.2, 70.7]; Table 7). The share of vendors with a positive valuation is slightly higher, at 69.4 per cent ([66.6, 72.2]).

What vendors value is protection, not productivity. Among vendors who chose the zone, 85 per cent cite protection from eviction or harassment and 60 per cent cite reduced stress and uncertainty, while fewer cite expected footfall or amenities (Figure 13, panel (b)). The value of formality here is largely insurance value. The formalisation literature predicts this: registration raises neither productivity nor earnings for most firms, and the welfare case rests instead on risk reduction and, for own-account entrepreneurs, on inclusion in social insurance (de Mel, McKenzie, and Woodruff 2013; Benhassine et al. 2018; Farias and Hsu Rocha 2026; Ulyssea 2020).

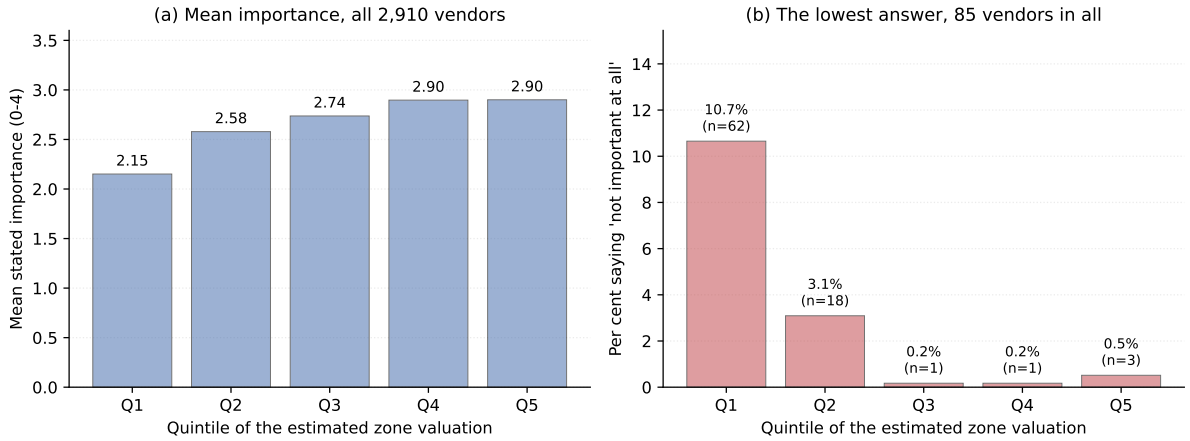
A stated attitude the model never saw confirms the estimated valuations. We elicited the importance of legal rights after the choice tasks, and after treatment for half the sample, so the item cannot enter the model. On a five-point scale, mean stated importance rises from 2.15 in the lowest quintile of the estimated valuation of a zone to 2.90 in the fourth quintile, and stays at 2.90 in the highest. The full cross-tabulation is far from independent ($\chi^2(16) = 323.8$), and every vendor in the estimation sample answered the item (Figure 9, panel (a)). The same pattern appears at the bottom of the scale. Vendors in the lowest quintile are about twenty times as likely as those in the highest to say legal rights do not matter at all: 10.7 per cent against 0.5 per cent. We report this alongside the means and do not lead with it, because only 85 of 2,910 vendors give that answer and three of the five quintiles contain fewer than five of them (panel (b)). Because the item is elicited after the choices, part of this alignment may reflect vendors rationalising choices already made. This concern applies equally to vendors who refused the zone in every round and to those who always accepted it.

The value of a zone depends on what a vendor expects of their peers. The figures above are evaluated at the sample-mean belief that 40 per cent of peers would also relocate. At a belief that no peers would move, the value of the zone falls from 2.60 to 0.52 in utility, or to about 0.9 per cent of sales through (5). We include the elicited belief in utility throughout. Without it, the assigned peer share is badly confounded: we constructed the assigned share by shifting each vendor's own belief by a randomised shock, so omitting the belief loads that construction onto the coefficient we use to bound coordination effects. The cost is that the average vendor's own expectation sits inside the intercept. Section 4.5.2 asks whether this dependence on beliefs reflects coordination among vendors.

4.3 Distance Consumes the Value of Formality

The second main finding is that relocation consumes the value of the legal right. Table 6 and Figure 10 report the maximum rent the average vendor would accept at each distance the design offered. It falls from 4.48 per cent of monthly sales at an unchanged commute to 2.09 per cent ([1.61, 3.33]) when the commute is one kilometre longer, and to 0.90 per

Figure 9: Stated Legal-Right Importance by Estimated Valuation



Legal-right importance was elicited after the choice tasks and, for the treated arm, after the information was given, so it is not a covariate in any model: this is a comparison the estimation never saw. Quintiles are of the posterior zone valuation under the main model, 582 vendors each. Panel (b) counts are on the bars because three of its five bars rest on one to three vendors; the contrast the text draws is between the first and the last. Chi-squared(16) = 323.7 on the full five-by-five table.

cent ([0.18, 1.46]) when it is three kilometres longer. Interpolating between the offered levels on either side, the valuation reaches zero at about 3.8 kilometres of additional commute ([2.94, 4.75]). At the five-kilometre move the design offered, the average vendor refuses the zone at any rent, including zero.[3]

Predicted take-up tells the same story. Table 7 and Figure 11 report predicted take-up at each of the seven zone designs the model can evaluate without extrapolation, with every other attribute held at each vendor's own status quo. At a free zone, take-up falls from 67.9 per cent with no change in commute to 61.9 per cent at one kilometre, 53.6 per cent at three kilometres and 45.5 per cent at five kilometres. Distance dominates rent over long moves. Charging two per cent of sales at an unchanged commute lowers take-up by 11 percentage points; a five-kilometre move lowers it by 22 points, twice as much, and the take-up intervals at the two designs do not overlap ([54.0, 59.4] against [42.0, 48.9]).

We do not decompose the distance response into a fixed cost of moving and a constant per-kilometre rate, because the design identifies that split only weakly. Offered distances cluster at the levels drawn, and the two terms move in opposite directions across samples while every quantity reported here stays put. The data support the cost of specific moves and the ladder above.[4]

The cost of distance reflects what the vendor gives up by leaving the site. Before seeing any choice profile, the average vendor expected only 14.9 per cent of regular customers to follow to a new site, and the median vendor just 5 per cent; 79 per cent expected lower customer volume there (Figure 12). Vendors' stated reasons point the same way. Figure

[3]No distance between three and five kilometres was ever offered, so the break-even figure interpolates linearly between those two levels. We report it to one decimal place for that reason.

[4]The experiment also offered commutes one kilometre *shorter* than the status quo, and we find no measurable effect of such offers. This does not establish that vendors place no value on a shorter commute. We floored reductions at the vendor's existing commute, so 986 of the 1,540 occasions on which a shorter commute was offered (64.0 per cent) delivered no change, and the mean realised reduction was 0.36 kilometres. The design does not identify the value of a shorter commute.

13 reports the reasons elicited after the final choice task, for staying in panel (a) and for relocating in panel (b). Among vendors who chose to stay, high rent at the new place (54 per cent) and distance from home (40 per cent) are the two most common reasons. Next come features of the incumbent spot: the time it takes to rebuild a customer base (39 per cent), attachment to the current area (30 per cent), and the risk that regular customers do not follow (24 per cent). For a firm trading on the street, location is closer to the production technology than to an overhead, because the customer base does not move with the firm.

The distance penalty does not look like generic status quo bias. Rent is the attribute on which status quo bias should bite hardest, yet a step for any change in rent is small and insignificant ($p = 0.24$), while the corresponding step for distance is significant ($p = 0.026$; Appendix A10). Expected competitors, expected customers and the peer share also show no significant step. The number of spots does ($p = 0.039$), so we cannot rule out some aversion to change on that attribute.

The ladder above is the average vendor's, and vendors differ widely around it. At an unchanged commute 69.4 per cent of vendors would accept the zone at some rent, and 44.4 per cent would accept a rent above eight per cent of sales, which the design never offered. The quartiles of the maximum acceptable rent are therefore taken over the 25.0 per cent of vendors who fall between the two: 1.67, 3.15 and 5.19 per cent of sales. After a one-kilometre move that group is 28.2 per cent of vendors, with quartiles of 1.38, 2.65 and 4.78. We do not extrapolate the top segment of the spline beyond the offered range, so we report no median over all vendors; among those with any finite maximum, 64 per cent lie above eight per cent of sales. The dispersion the design identifies without any such restriction is the share of vendors who accept at each offered rent, in Section 4.5.1.

Figure 10: Maximum Acceptable Rent by Distance

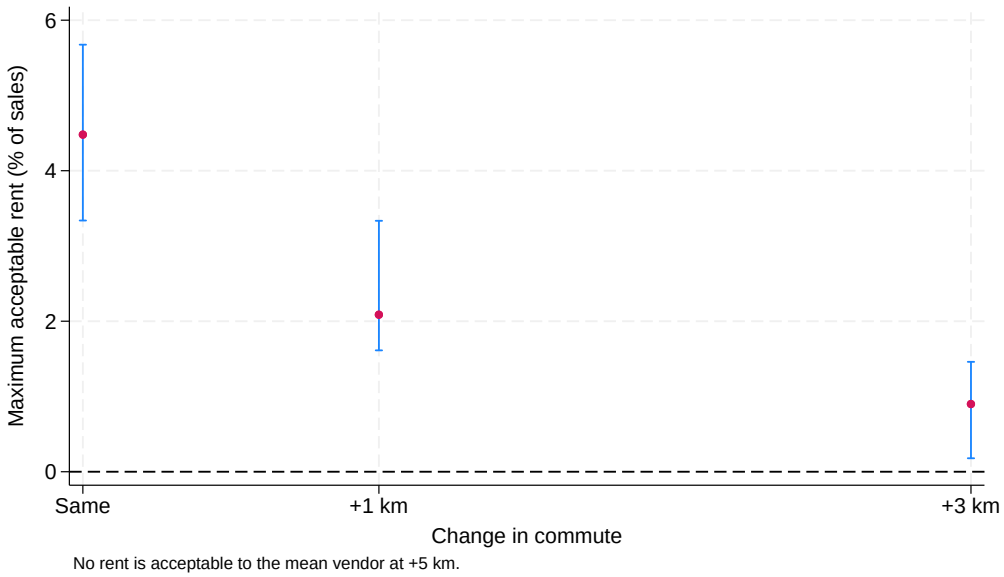


Table 6: The Value of a Zone and the Cost of Distance

	Estimate	95% CI
Panel A. Maximum acceptable rent, per cent of monthly sales		
At an unchanged commute	4.48	[3.34, 5.68]
after a +1 km move	2.09	[1.61, 3.33]
after a +3 km move	0.90	[0.18, 1.46]
after a +5 km move	none	
At an unchanged commute, with information	5.83	[4.46, 7.66]
Break-even distance (km)	3.79	[2.94, 4.75]
Panel B. Trade-offs, in kilometres of commute		
Rent, per point below 1 per cent	0.99	[0.43, 1.69]
Rent, per point between 1 and 2 per cent	1.84	[1.40, 2.59]
Rent, per point above 2 per cent	0.64	[0.47, 0.90]
Information treatment	0.86	[0.39, 1.56]
Peer share, per 10 percentage points	0.00	[-0.08, 0.08]
Panel C. The zone in utility		
Value at the mean peer belief	2.60	[2.12, 3.07]
at a belief that no peers move	0.52	[0.13, 0.90]
with information	3.10	[2.54, 3.78]
Step ratio, $\beta(+1)/\beta(+3)$	0.43	[0.31, 0.56]

Notes: Mixed logit with a random intercept and a random per-kilometre slope, 2,910 vendors and 14,550 choice occasions. Intervals are a 200-replication marketplace pairs bootstrap. Rent is a percentage of each vendor's own monthly sales; one point is INR 607 at median sales. At a +5 km move there is no rent, not even zero, at which the average vendor accepts, which is the same fact as the break-even distance. The break-even distance interpolates between the +3 and +5 km levels, the only two the design offered either side of it.

Table 7: Predicted Take-up across the Design Ladder

Design	Δ km	Rent (%)	Take-up (%)	95% CI	With information	Difference
D0	0	0	67.9	[65.2, 70.7]	71.1	3.2
D1	0	1	64.2	[61.8, 66.9]	67.5	3.3
D2	0	2	56.7	[54.0, 59.4]	60.3	3.5
D3	1	0	61.9	[58.7, 64.8]	65.3	3.4
D4	1	1	58.0	[54.4, 60.8]	61.4	3.5
D5	3	0	53.6	[50.7, 57.1]	56.9	3.3
D6	5	0	45.5	[42.0, 48.9]	48.5	3.0

Notes: Predicted share of vendors relocating at each zone design, averaging over the estimated distribution of vendor valuations, with every attribute other than distance and rent held at each vendor's own status quo and the peer share at each vendor's own elicited belief. Rent is a percentage of own monthly sales. The last column is the effect of the information treatment at that design. These are the seven designs the model can evaluate without extrapolating beyond the offered levels.

Figure 11: Predicted Take-up across the Design Ladder

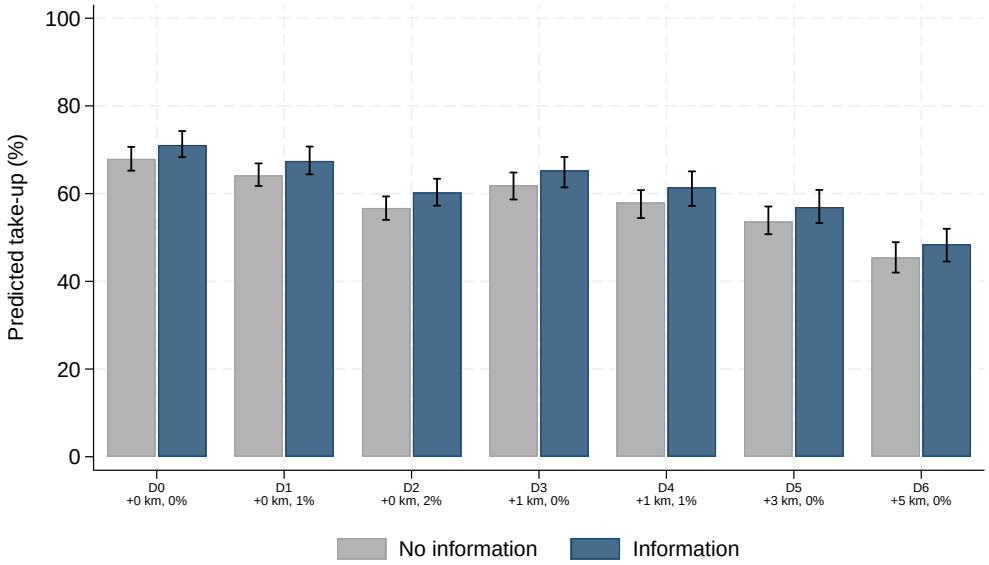
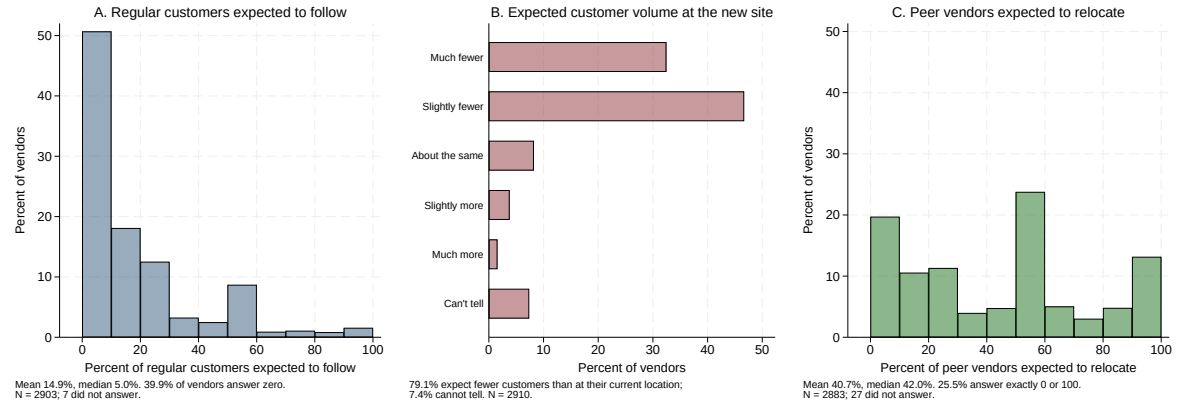
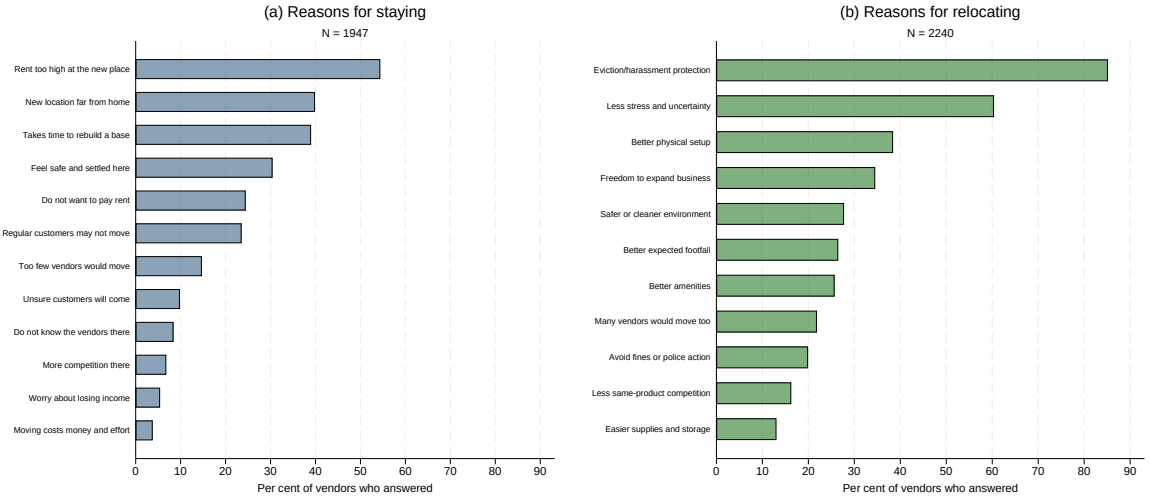


Figure 12: Pre-Choice Belief Elicitations



All three questions were asked before the first choice profile was shown:
they are items 1-3 of the choice module and the cards are item 4.
Estimation sample, one observation per vendor.

Figure 13: Stated Reasons for Staying and for Relocating



Notes: Reasons were elicited after the final choice task and, for the treated arm, after the information was given. They are descriptive only: no quantity estimated in this paper uses them. Each question was asked of every vendor who made that choice at least once across all six choice tasks, the practice round included, on the survey sample of 2,922 vendors. Panel (a) is the 1,947 vendors who chose to stay at least once and panel (b) the 2,240 who chose to relocate at least once, so the denominators differ, a vendor who did both appears in both panels, and neither reconciles with the 748 and 1,021 vendors of Section 4.1.2, which count the five analytical rounds on the 2,910-vendor estimation sample. Respondents could give more than one reason, so shares do not sum to 100. "Other" is omitted: 1 respondent for staying and 9 for relocating.

4.4 Information Raises the Value of Relocating

The third main finding is that information about the Act raises the value of relocating without changing any of the trade-offs between the zone's attributes (Section 3.3). Table 5 reports the Treat × Post coefficient on the zone intercept. In willingness-to-pay terms, the treatment is worth 1.35 per cent of monthly sales ([0.62, 2.25]), or 0.86 kilometres of commute ([0.39, 1.56]). It raises the maximum rent the average vendor would accept at an unchanged commute from 4.48 to 5.83 per cent of sales (Table 6), and lifts predicted take-up by about 3.3 percentage points at each of the seven designs in Table 7.

At baseline, only 1.3 per cent of vendors could name the Act and only 7.7 per cent were aware of any such policy (Figure 5). The effect is therefore consistent with an information channel: most vendors did not know the policy existed or was being enforced elsewhere, and learning this raises the value of relocating. The treatment also tells vendors that they will lose their current site once the policy is enforced, so part of the effect may come from a higher expected cost of staying. Because the treatment combines general policy information with a specific example of enforcement, we cannot separately identify whether the effect operates through awareness or through the credibility of a concrete example. The low baseline awareness also means the estimate is likely an upper bound on the return to a marginal information campaign.

Information shifts the level of the valuation, not its gradient in distance. Because the treatment changes no trade-off, it lifts take-up by about the same amount at an unchanged commute as after a five-kilometre move. A 3-point gain does not offset the 22 points that

a five-kilometre move costs (Section 4.3). Information can raise the value of a vending zone, but it cannot substitute for addressing where the zone is.

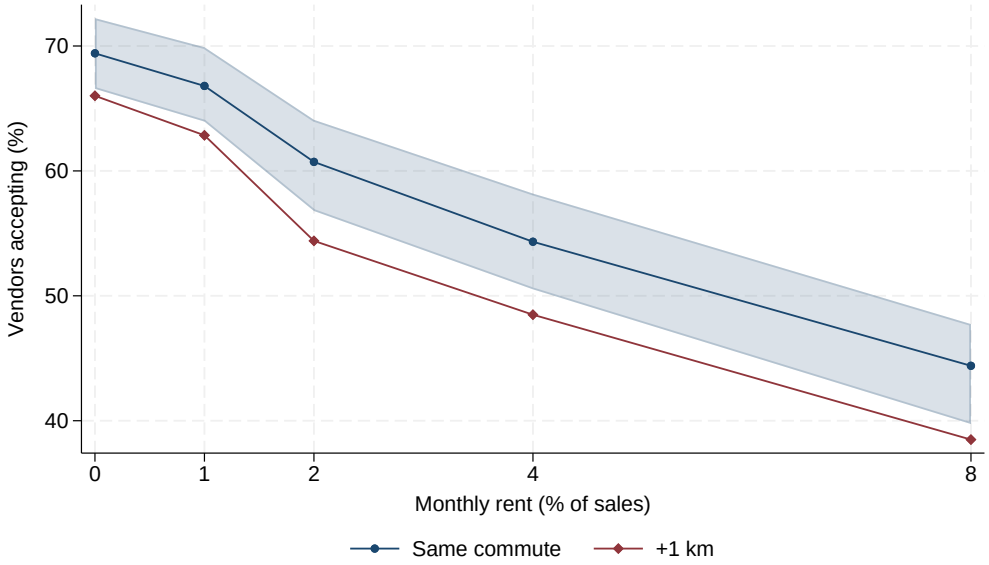
4.5 Secondary Results

Four further results qualify the main findings. Rent matters but is bounded; we find no detectable response to the share of peers expected to relocate, within limits the design sets; the number of spots in a zone has no detectable effect; and the dispersion of valuations limits what targeting can achieve.

4.5.1 Rent

Rent carries disutility from the first percentage point of sales. A percentage point of rent is worth about 1.0 kilometre of commute below one per cent ([0.43, 1.69]), 1.8 kilometres between one and two per cent ([1.40, 2.59]) and 0.6 kilometres above two per cent ([0.47, 0.90]). Figure 14 shows the share of vendors with a positive valuation at each rent the experiment offered. At an unchanged commute, this share is 69.4 per cent at no rent, 66.8 per cent at one per cent of sales, 60.7 per cent at two, 54.3 per cent at four and 44.4 per cent at eight. After a one-kilometre move, the share is lower at every rent. Rent matters because it is a new obligation: 91 per cent of vendors pay nothing for their current pitch. Rent is therefore a real instrument, but over long moves it cannot offset the cost of distance (Section 4.3).

Figure 14: Share of Vendors with a Positive Valuation, by Rent Level, at an Unchanged Commute and after One Kilometre



4.5.2 Coordination among Vendors

Coordination failure is the explanation most often advanced for unoccupied formal space (Cooper and John 1988; Brock and Durlauf 2001). We find no detectable effect of the expected share of relocating peers on the choice to relocate. The estimate is −0.0005 kilometres of commute per ten percentage points, with a 95% confidence interval of

$[-0.077, 0.077]$. In money, it is -0.0007 per cent of sales ($[-0.122, 0.110]$). Setting the expected peer share to 100 per cent, so that the whole marketplace relocates together, changes predicted take-up by no more than 0.3 points at any design in Table 7.

The cross-sectional data point the other way. Vendors who expect more peers to relocate are much more willing to relocate themselves, and Section 4.2 shows that the value of a zone depends on this belief. But pessimism about peers also travels with a lower valuation of the zone itself, so the observational association confounds the effect of peer behaviour with whatever else separates optimistic vendors from pessimistic ones: the simultaneity problem of Section 3. When we assign the share instead of eliciting it, take-up does not respond. Few vendors who chose to stay cite too few peers moving as a reason (Figure 13, panel (a)).[5]

Three features of the design limit how far this null extends. First, it holds over a narrower range of peer shares than the design nominally spans. We censored the assigned share at zero and 100, and 35.5 per cent of occasions sit at one of these bounds, so a vendor who already expected ten per cent of peers to move received at most a ten-point reduction. The mean shock the design delivered is 12.8 percentage points against a nominal 15.6. Over that delivered range the interval rules out an effect larger than about 0.08 kilometres of commute per ten percentage points of peer share.

Second, the design varies an anonymous share of relocating peers. It does not tell vendors which peers would move, and a vendor may care about specific neighbours, suppliers or relatives more than about the marketplace as a whole. Third, the peer share is a stated expectation in a hypothetical choice, while coordination failure in the field may operate through what vendors observe others actually doing. The result therefore rules out a large response to an anonymous, stated peer share. It does not rule out coordination failure in every form.

4.5.3 Zone Size

The number of spots in the zone has no detectable effect on relocation across the design range of 20 to 160 spots (Appendix Table A4). One reading is that, conditional on expected competitors and the expected share of peers relocating, additional spots represent room for entrants who are not yet present, which an established vendor has no reason to value. The estimate is too small for us to rely on that interpretation. It does rule out the simplest policy response to an empty zone, which is to build a larger one. This attribute is the number of spots, not the floor area of each spot, which the choice tasks hold fixed. Section 4.6 discusses spot dimensions.

4.5.4 Heterogeneity and Targeting

Because the value of a zone is so dispersed (Section 4.1.2), the terms of the offer determine which vendors a zone draws, and a municipality cannot identify those vendors in advance.

[5] Reported ties within the marketplace do not predict the valuation of a zone either. We asked vendors how many others in their market they drink tea with, could ask to mind their stall, share business information with, could borrow from, would lend to, and are related to. These counts cover relations with other vendors in the current marketplace, so they do not speak to attachment to customers, to the location itself or to ties elsewhere in the city. Appendix A9 reports the estimates.

Grouping vendors by what the 27 observed characteristics predict about them spreads take-up at the best design from 49.9 to 86.0 per cent, a range of 36 points. Grouping them by their estimated valuation, which uses their own choices, spreads it from 7.5 to 99.9 per cent, a range of 92 points. The difference is the part of the variation a survey cannot see. Part of the heterogeneity is predictable and large in take-up terms, but it is the smaller part, so a programme cannot target the vendors most likely to relocate with any precision.

The share of vendors whose choices respond to the terms on offer depends on how the question is posed, so we report two measures. Across the seven designs in Table 7, the predicted relocation probability of 34.9 per cent of vendors ([32.0, 37.3]) changes by more than 20 percentage points between the best and worst design. Across the full grid of levels the design drew, which includes combinations no municipality would offer, the figure is 57.8 per cent ([49.6, 64.8]). We take the first as the headline, because a policy could act on it. Both are lower bounds, for the reason given in the footnote to Section 4.1.2.

4.6 Space Constraint and the Formalisation Gap

Our experiment cannot price one dimension of zone design: the space allocated to each vendor. The choice tasks hold operating scale fixed by construction. Under the Bihar implementation of the Act, each spot measures 6 by 4 or 5 by 3 feet. In our sample, 85 per cent of vendors operate from a stall or cart, not a floor display (Table 1), and many sell perishable produce or cooked food that requires storage, preparation space and customer-facing display. We do not measure current footprints, so the comparison rests on the stated spot dimensions and the observed distribution of stall types, not on a direct estimate.

Spot size offers one explanation for the gap between the willingness to relocate we estimate and the near-zero take-up observed. Our choice tasks tell vendors that the zone differs from their current spot only in the attributes shown, and that storage capacity is unchanged. The vendors who accept a zone at an unchanged commute are therefore vendors who would relocate if a zone accommodated them at their current scale. The zones as built may not. If so, even vendors willing to move on every other margin face a constraint that rent subsidies, information campaigns or coordination mechanisms cannot relax without first addressing the space allocated per vendor. This is a second spatial margin alongside the siting of the zone. Two other explanations for the gap remain: the built zones may lie far from where vendors live and trade, and stated choices may overstate actual ones.

5 Conclusion

Place-based formalisation fails on the terms of the offer, not on the value of what is offered. Vendors do not reject the legal right itself. If a municipality designates an existing site as a zone, so that no commute changes, the average vendor's willingness to pay for a spot is 4.5 per cent of monthly sales, and predicted take-up of a free spot is 68 per cent. Among vendors who choose a zone, 85 per cent cite protection from eviction or harassment as the reason. Relocation consumes that value. Willingness to pay falls from 4.5 per cent of sales at an unchanged commute to 2.1 per cent one kilometre away and 0.9 per cent three kilometres away, and reaches zero at about 3.8 kilometres, beyond which

vendors refuse a zone even when the spot is free. Information about the Act raises the value of relocating, lifting take-up by about 3 percentage points at every design, but it does not change how vendors trade off distance against the zone's other attributes, so it cannot offset a distant zone. The other explanations account for less. Rent matters, but a five-kilometre move lowers take-up twice as much as a rent of two per cent of sales. The number of spots has no detectable effect, so building larger zones is not a remedy. And we find no detectable response to the share of peers a vendor expects to relocate, within limits the design sets. Four margins decide whether a zone fills: whether the vendor has to move, how far, what the spot costs, and how much space it offers.

5.1 Implications for design

Our estimates price the first three of these margins and bound the fourth.

Designate in place where possible. Where an existing vending site can itself be designated as a zone, the legal right is delivered without the cost of moving. On our estimates, this is the best option available. It is also the option the municipality has the least room to choose. The sites vendors occupy are the sites whose congestion the policy exists to relieve, and 90 per cent of vendors in our sample trade along main roads or streets, with the remaining 10 per cent on pedestrian walkways. In-situ designation will therefore be feasible in some marketplaces and not in others. Our estimates price what displacement costs the vendor. They do not price what leaving vendors in place costs the city, and the comparison that would settle the choice at any particular site requires both.

Where displacement is unavoidable, keep the move short. Take-up at a free zone falls from 68 per cent at an unchanged commute to 62 per cent one kilometre away, 54 per cent three kilometres away and 46 per cent five kilometres away. After a one-kilometre move, the average vendor's willingness to pay is 2.1 per cent of sales, against 4.5 per cent for the same zone designated in place. Even a short move therefore costs the municipality about 2.4 percentage points of monthly rent for the average vendor. Over long moves, rent cannot buy the distance back: waiving a rent of two per cent of sales recovers 11 points of take-up, while a five-kilometre move costs 22.

Set rent with care. Rent carries disutility from the first percentage point of sales, so no band of rent is free. A percentage point of rent is worth about 1.0, 1.8 and 0.6 kilometres of commute in the three segments of the spline. Rent is a real instrument, but each percentage point comes out of a valuation that relocation has already reduced.

Make the spots fit the vendors. The spot dimensions fixed by the Bihar rules appear smaller than the space most vendors need to run their operations. Our choice tasks hold operating scale constant by construction (Section 4.6), so stated willingness to relocate assumes a spot that fits. Larger spots may therefore be needed to turn stated willingness into take-up, even at a zone designated in place. Our design cannot measure how much this matters.

Information helps, within limits. Information provision raises the value of a zone by about 1.4 per cent of monthly sales, or 0.9 kilometres of commute, and lifts take-up by

about 3 percentage points at every design we evaluate, without changing any trade-off. It is cheap to deliver, and it moves take-up in the right direction. But because it shifts the level of the valuation and not its gradient in distance, it cannot substitute for siting. Only 1.3 per cent of vendors could name the Street Vendors Act at baseline, and only 7.7 per cent were aware of any such policy, so the estimate is likely an upper bound on the return to a marginal campaign.

Coordination mechanisms are unlikely to be enough. The standard prescription is that vendors must be moved together so that none moves first. Within the range our design covers, the data do not support it: assigning a vendor the expectation that the entire marketplace will relocate changes predicted take-up by no more than 0.3 points at any design. This result rules out a large response to an anonymous, stated peer share. It does not rule out coordination failure that operates through specific peers or through what vendors observe others doing (Section 5.3).

Enforcement shifts the cost to vendors. Enforcement is the instrument the formalisation literature identifies as most effective at inducing formality (Andrade, Bruhn, and McKenzie 2016; Ulyssea 2020), and it carries a corollary here. If zone spots are too small to house existing operations, raising the cost of informality would not move these vendors into the formal economy. It would make informality more expensive without opening a viable alternative. Singapore, the one case we found in which the empty-zone equilibrium was decisively broken, did exactly this: it withdrew street hawker licences instead of improving the zones (Appendix A1), and the cost of any mismatch between what vendors needed and what the zones offered fell on the vendors.

Siting, not targeting, is the policy variable. Vendors differ enormously in what a zone is worth to them. The 27 characteristics we observe, including earnings, education, tenure and exposure to eviction, generate 15.5 per cent of that variation and reduce the unexplained part by only 6.1 per cent. A programme therefore cannot be aimed with any precision at the vendors most likely to take it up. What a municipality can vary is the terms of the offer.

Taken together, the estimates locate the failure of these programmes in a constraint, not a mistake. Zones sit empty not because the state has failed to offer formality or because vendors do not want it. They sit empty because the only instrument available for conferring legal status on a firm without premises requires moving it, and moving it consumes most of what the legal status is worth. The state is not absent here, and the firms are not myopic. What binds is that the site where a vendor is most willing to be formalised is the site where the city is least willing to leave them.

5.2 Transition costs and the long run

Our estimates price the decision to relocate at a point in time. That decision bundles two things a policymaker would want to separate: the cost of moving and the cost of being somewhere else. Part of the cost we measure is plausibly transitional. Among vendors who chose to stay, 39 per cent cite the time needed to rebuild a customer base, and smaller shares cite the money and effort of setting up again (Figure 13). Such costs are temporary by construction, so part of what we attribute to leaving the site may be

the disruption of moving, not a permanent disadvantage of the destination. If so, the short-run cost of relocation exceeds its long-run cost, and a design that looks unattractive on our estimates could still leave vendors better off once adjustment is complete.

Several channels would push in that direction. Secure tenure lengthens the horizon over which investing in a spot pays off (Harari and Wong 2026). Recognition brings eligibility for collateral-free lending under PM SVANidhi. Destinations need not stay as they begin (Franklin 2025; Agness and Getahun 2024). And a lawful spot removes a recurring source of loss, not a one-off one. The evidence also cuts the other way. Relocated households in Mumbai were no better off than those who stayed (Barnhardt, Field, and Pande 2017), and formalisation raises neither productivity nor earnings for most firms (de Mel, McKenzie, and Woodruff 2013; Farias and Hsu Rocha 2026). We observe the choice, not what follows from it.

Whichever way that evidence settles, our estimates point to an asymmetry that design should take seriously. The vendor bears the transition cost immediately and with certainty; the long-run gain is uncertain and deferred. For a firm with the profit levels reported in Table 1, that asymmetry is not a matter of patience but of whether the business survives the intervening months. This is an argument for compensating the move instead of assuming it will pay for itself, through a rent holiday while the vendor rebuilds a customer base or through the working capital that PM SVANidhi already provides. A rent holiday suits this purpose, because it covers a cost that is itself temporary.

5.3 Limitations

We recover these estimates from stated choices because the setting permits nothing else. Ninety-nine per cent of vendors have never changed marketplace, and the zones that exist are empty. There is no revealed variation in the terms of relocation, and a revealed-preference design here would recover only the preferences of vendors who had already moved. The experiment supplies variation the field does not, at the cost that the choices are hypothetical. The design personalises every attribute to each vendor's verified circumstances, and we introduced the choices as informing PMC policy. Both steps narrow the gap between stated and actual behaviour, but neither guarantees the level of the estimates. Randomisation identifies the comparisons across designs on which our policy conclusions rest. Levels, such as the 4.5 per cent willingness to pay and the 68 per cent take-up, carry this caveat.

Because the choice tasks hold operating scale constant, we bound the space allocated per vendor instead of pricing it. The Bihar spot dimensions appear smaller than the operations most vendors run, which is consistent with the observed pattern of take-up. But we cannot say what a vendor would pay to avoid shrinking, or how that cost ranks against distance and rent. Our account of the gap between stated willingness and observed take-up is therefore an interpretation the evidence supports, not a quantity we estimate. It rests on the stated spot dimensions and the observed distribution of stall types, not on measured footprints. Six attributes also do not exhaust the design of a vending zone. We hold constant tenure duration, enforcement inside the zone, infrastructure and eligibility for schemes such as PM SVANidhi, so the estimates price what we vary, conditional on the rest of the offer being as vendors imagine it.

Two features of the design limit particular results. First, the experiment identifies the cost of a longer commute but not the value of a shorter one, because we floored reductions at each vendor's existing commute and most such offers delivered no change. Second, the coordination result is limited in three ways. It holds only over the range of peer shares the randomised shock actually delivered, which is narrower than the design's nominal range because we censored the assigned share at zero and 100. It concerns an anonymous share of relocating peers: reported ties within a marketplace are few relative to its size (Appendix A9), and the design does not separate a vendor's response to that share from a response to knowing which particular vendors are moving. And it concerns a stated expectation in a hypothetical choice, while coordination in the field may operate through what vendors see others actually do.

The data reject one restriction in our specification. Allowing the cost of changing site at all to vary across vendors, as well as the cost of each kilometre, improves fit on every criterion we examine. It changes the maximum rent at an unchanged commute, the break-even distance and the value of information by less than one per cent, and moves the intermediate rungs of the ladder by at most eight per cent. It does, however, more than double the dispersion of the distance response over short moves (Appendix A8). Statements about how widely vendors differ at short distances are therefore lower bounds, and we report the more conservative model.

We price the relocation decision, not its consequences. Whether vendors who relocate earn more, are evicted less or are better off is a separate question that preferences cannot settle. We also do not measure the congestion costs that motivate relocation, so these estimates bound what displacement costs the vendor without establishing when displacement is nonetheless warranted. The setting is a single city. The instrument is common across the developing world (Appendix A1), but the spot dimensions, rent levels and commuter-led demand are Patna's. We elicited the stated reasons in Figure 13 after the final choice round, after treatment for the information arm, and the beliefs in Figure 12 before vendors saw any choice profile.

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Appendix

A1 Relocation Policies Around the World

Figure A1: News Articles from Around the World



The formalisation model we study in Patna, in which vendors receive legally protected spots in a designated zone in exchange for relocating, is not peculiar to Indian urban policy. Table A1 documents 54 policy instruments in 27 countries across six regions. All but one date from 2001 to 2026. The exception is Singapore's hawker resettlement programme of the 1970s and 1980s, the historical template for much of the contemporary model. Just over 70 per cent of the entries designate or construct a physical site to which vendors are expected to move, which is the margin our experiment evaluates. The approach spans low-income countries with very large informal sectors (Tanzania, Uganda, Bangladesh), middle-income countries undergoing rapid redevelopment (Indonesia, Vietnam, Peru, Egypt), and high-income countries where vending is a small but politically salient share of employment (Singapore, the United States). Governments of every political stripe use it. The approach is also durable: R. Bromley (2000) reviewed the same family of instruments a quarter of a century ago, and the design has changed little since.

Table A1: Street vendor formalisation, relocation and vending zone policies

Country and city	Policy or programme	Type
Panel A: South Asia		
India	National Policy on Urban Street Vendors (2004, rev. 2009)	L
India	Street Vendors (Protection of Livelihood and Regulation of Street Vending) Act (2014)	L, Z, R
India	PM SVANidhi micro-credit scheme (2020)	R
India, Bhubaneswar	Participatory vending zone regularisation (from 2007)	Z, R
Bangladesh, Dhaka	City Hawker Management Policy (2026)	L, Z, R

Continued on next page

Table A1, continued

Country and city	Policy or programme	Type
Bangladesh, Dhaka	DNCC temporary markets on six open fields (2026)	Z
Nepal, Kathmandu	KMC space allocation proposals and clearance (2010s)	E
Panel B: Southeast and East Asia		
Thailand, Bangkok	BMA vending ordinance (2002) and Designated Areas Directive (2005)	L, Z, R
Thailand, Bangkok	Return the Footpath to Pedestrians (from 2014) (Boonjubun 2017)	E, Z
Thailand, Bangkok	Revised BMA rules and hawker centre plan (2020s)	Z, R
Indonesia, Surakarta	Notoharjo and Panggung Rejo relocation (2006); Regulation No. 3 (2008) (Permatasari et al. 2014)	Z
Indonesia, Yogyakarta	Malioboro to Teras Malioboro relocation (2022)	Z
Indonesia, Jakarta	Relocation to designated lots (from 2015)	Z, E
Vietnam, Hanoi	Vending ban, 62 streets and 48 public spaces (2008) (Eidse, Turner, and Oswin 2016)	E
Vietnam, Ho Chi Minh City	Civilized Streets (2013 to 2015); sidewalk reclamation (2017)	E
Vietnam, Ho Chi Minh City	Nguyen Van Chiem and Bach Tung Diep food zones (2017)	Z, R
China	Street stall economy, State Council measures (2020) (Song 2020)	L, Z
China, Shenzhen	Designated vending areas (2023)	Z
Singapore	Hawker resettlement programme (1970s to 1986)	Z, R
Singapore	Hawker Centres Upgrading Programme (2001); SEHC model (2015)	Z
Philippines, Quezon City	Revised Market Code, Ordinance SP-2459 (2015)	L, Z, R
Philippines, Manila	Ordinance No. 8349 on designated vending areas	L, Z
Philippines	House Bill 10724, proposed national framework (2024)	L, Z, R
Panel C: Sub-Saharan Africa		
South Africa, Durban	Informal Economy Policy and Warwick Junction upgrade	L, Z
South Africa, Johannesburg	Operation Clean Sweep (2013) (Bénit-Gbaffou 2016)	E, Z
Tanzania	Vendor recognition (2016); Wamachinga ID cards (2018) (Msoka and Makundi 2023)	R
Tanzania, Dar es Salaam	Machinga Complex and Kariakoo pitch allocation (from 2017)	Z
Rwanda, Kigali	Vending ban and mini-market programme (2017, 2022) (Gisanabagabo and Masengesho 2024)	E, Z
Uganda, Kampala	KCCA gazetted markets directive (2022)	E, Z

Continued on next page

Table A1, continued

Country and city	Policy or programme	Type
Kenya, Nairobi	Pop-Up Markets and Street Vending Act (2019)	L, Z, R
Kenya, Nairobi	Mwariro and Karandini market relocation (2017 to 2019)	Z
Ghana, Accra	AMA decongestion by-laws (2011) and clearance exercises (Crentsil and Owusu 2018; Steel, Ujoranyi, and Owusu 2014)	E, Z
Zimbabwe	Operation Murambatsvina (2005) (Musoni 2010)	E, Z
Zimbabwe, Harare	Designated vending sites directive (2015)	E, Z
Liberia, Monrovia	Memorandum of Understanding with the City Corporation (2018)	L, R
Senegal, Dakar	Petersen Market eviction and PASI negotiations (2023)	E
Panel D: Middle East and North Africa		
Egypt, Cairo	Relocation of downtown vendors to Torgoman (2014)	E, Z
Egypt, Cairo	Six alternative public markets, incl. El Tunisi, El Turjuman and Ahmed Helmy (Mohammed Darwish 2023)	Z
Egypt, Cairo	Al-Tounsi Market relocation, UNESCO management plan (2010 to 2014)	Z
Panel E: Latin America		
Mexico, Mexico City	Programa Integral de Rehabilitacion del Centro Historico (2007) (Crossa 2009)	Z, E
Peru, Lima	La Parada closure and Santa Anita relocation (2012) (Coletto, Jaber, and Vanhellemont 2021)	Z, E
Colombia, Bogota	IPES REDEP kiosk network (Donovan 2008)	Z, R
Colombia, Bogota	Politica Publica de Vendedores Informales 2023 to 2035; RIVI register	L, R
Colombia, Bogota	Public space decree, confianza legitima protocol (2026)	L, Z
Ecuador, Quito	Relocation to eleven Centros Comerciales de Ahorro (2003) (Middleton 2003; R. D. F. Bromley 1998)	Z
Brazil	Microempreendedor Individual, Complementary Law 128 (2008)	L, R
Brazil, Sao Paulo	To Legal, Termo de Permissao de Uso for ambulantes	R
Chile, Santiago	Ordenanza No. 127 on comercio estacionado (rev. 2026)	L, R
Paraguay, Ciudad del Este	Municipal formalisation project	Z, R
Panel F: North America		
United States, California	Safe Sidewalk Vending Act, SB 946 (2018)	L, R
United States, Los Angeles	City and County Sidewalk Vending Ordinances (from 2018)	L, R

Continued on next page

Table A1, continued

Country and city	Policy or programme	Type
United States, California	SB 972, retail food code reform (2022)	L, R
United States, New York City	Local Law 18, permit expansion and enforcement office (2021)	L, R
United States, New York City	Advisory Board recommendations on designated spaces (2022)	Z

Notes: Non-exhaustive. Street vending is regulated mainly at municipal level, so the full universe runs to thousands of by-laws, most undocumented in academic or English-language sources. This is a curated sample weighted towards cases with an identifiable relocation or zone-creation component and a citable source. Academic treatments are cited where they exist; otherwise the policy name links to its primary source. Types: L, legislation or policy framework; Z, designation or construction of vending zones or market facilities; R, registration or licensing of vendors; E, clearance paired with an offer of an alternative site. Entries may carry more than one code. Spanish and Portuguese titles are given without diacritics.

Several of these cases closely match the Patna setting: a government built a purpose-built facility, and vendors under-occupied it for a documented reason. Mohammed Darwish (2023) evaluates six alternative public markets in Cairo and identifies unprofitable location as the main reason vendors rejected them, together with stall designs that did not meet vendors' operating requirements. In Surakarta, the Notoharjo market was well attended at first but drew persistent complaints of low footfall as its facilities deteriorated (Permatasari et al. 2014). In Lima, larger wholesalers moved to the new Santa Anita facility, while smaller traders, who could not afford the rents, stayed put (Coletto, Jaber, and Vanhellemont 2021). In Kampala, a survey after the 2022 directive to trade only from gazetted markets found average daily earnings of UGX 50,000 among street traders against UGX 7,000 among traders in the zoned markets, and 85 per cent of relocated traders reported lower income (Gateway Research Centre 2022). In Nairobi, the Mwariro market was built with about 350 stalls for some 7,000 traders who were to leave the central business district (PesaCheck, 2018), and its stalls remained unoccupied years after completion (*Daily Nation*, 2024). These cases match the margins our study examines: distance and forgone footfall, rent, and allocated space. Our design prices the first two and bounds the third. The qualitative literature recognises the pattern. R. D. F. Bromley and Mackie (2009) show how stall allocation rules and pricing in new Latin American municipal markets excluded the poorest traders, and Roever and Skinner (2016) report vendor complaints about the viability of relocation sites in Lima and Mexico City. This literature cannot quantify how much of the rejection is due to distance, rent or the spot itself, because these move together across cases and observational data cannot separate them. Separating the cost of distance from the cost of rent is our contribution.

Singapore offers an instructive contrast, because it is one of the few documented cases in which the empty-zone equilibrium broke. In 1975, one in six stalls in the new hawker centres stood empty while some 15,000 hawkers still traded on the street. The records give three reasons for their reluctance: street spots carried no rent, hawkers had established clienteles they feared losing, and a street stall could be enlarged simply by adding tables and chairs (National Archives of Singapore 2026). These are the same three margins we study, half a century earlier and on another continent. The government did not reverse the equilibrium by improving the zones. It removed the alternative. Licensed hawkers had

to occupy their allocated stalls within two weeks or face a month's suspension, and the government stopped issuing new street hawker licences in 1978. Street hawker numbers fell to about 4,000 within five years and to about 1,000 by 1983, and the last street hawker was relocated in 1986 (National Archives of Singapore 2026).

A2 Certification and Enforcement Exposure

Section 2 reports that certification under the SVA has reached a fifth of our sample without protecting the sites vendors occupy. Table A2 gives the comparison behind that statement.

Certified vendors report more enforcement exposure than uncertified vendors, not less. The gaps are large: 23 percentage points on harassment and 16 on eviction or confiscation, both significant at well under one per cent. Certification is not randomly assigned, so neither figure is the effect of holding a certificate. Two selection routes could produce the sign. The first is between marketplaces: PMC surveyed and certified vendors where it was clearing pitches, and the most congested marketplaces are both the likeliest to be surveyed and the likeliest to be cleared. The second is between vendors within a marketplace: a vendor who has been harassed or evicted has the strongest reason to obtain a certificate.

The two columns separate them. Marketplace fixed effects remove the first route entirely, and they cut the gaps by roughly a third, from 23.2 to 14.6 points on harassment and from 15.8 to 11.0 on eviction. Most of each gap is therefore between vendors trading in the same marketplace. The last row is a further check on the between-marketplace account: certified vendors are no more likely than uncertified ones to be in a marketplace where eviction drives occur monthly or more often, and the point estimate is slightly negative. We read the within-marketplace gap as the second route, vendors seeking certification after enforcement rather than certification inviting it.

Neither reading changes what the result means for the paper. The right the Act confers is a legally protected spot inside a vending zone, so a certificate held by a vendor who has not moved carries no protection for the pitch they still occupy. That is the sense in which the Act removes the fixed costs of registration while leaving the spatial condition untouched, and it is why low take-up points to the terms of relocation rather than to the cost of registering.

Table A2: Certification and Enforcement Exposure

	Share reporting (%)		Difference (pp)		
	Certified	Uncertified	Raw	Within mkt.	N
Harassed by officials or police	61.0	37.8	23.2***	14.6***	2870
			(3.0)	(2.7)	
Evicted or goods confiscated	45.6	29.8	15.8***	11.0***	2909
			(3.2)	(3.1)	
Either	66.7	46.8	19.9***	11.5***	2869
			(3.2)	(3.1)	
Marketplace drives monthly or more	68.0	70.3	-2.4	-4.6	2916
			(3.2)	(2.9)	

Notes: Survey sample of 2,922 vendors in 206 marketplaces; 20.4 per cent hold a certificate of vending. Each row is one outcome. The raw column regresses the outcome on the certificate indicator; the within-marketplace column adds marketplace fixed effects, so it compares certified and uncertified vendors trading in the same place. Standard errors clustered at the marketplace in parentheses. Refusals and don't-know answers are set to missing rather than to zero, so N varies by row; the *either* row is missing only where both items are. The last row is the vendor's report of how often eviction drives occur in their marketplace, at the monthly-or-more threshold the text quotes in Section 2. Certification is not randomly assigned, so these are associations and not effects of holding a certificate. Marketplace fixed effects cut the gaps by roughly a third (37, 31 and 42 per cent of the raw gap across the first three rows), so part of the difference is which marketplaces PMC surveyed, but most of it is between vendors trading side by side; and certified vendors are not in marketplaces with more frequent drives, which is what the last row shows. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A3 Representativeness

To assess representativeness, we compare survey respondents with vendors who were recruited but did not complete the survey. Table A3 presents the comparison. Respondents differ from non-respondents on two observable characteristics. Respondents are more likely to sell produce (59.2 against 49.5 per cent), and therefore less likely to sell cooked food and drinks. They are also slightly less likely to be male (88.3 against 91.0 per cent). Age, trading location and the use of stalls or carts are balanced across the two groups. Two features of the data collection explain these patterns. First, street food vendors must attend to preparation and service more continuously than produce vendors, which leaves them less free time for a 45-minute survey. Second, in businesses run by a husband and wife, recruited husbands often delegated the main survey to their wives, which lowers the share of male respondents. The differences are modest and reflect operational and household constraints on participation. Respondents and non-respondents do not differ in trading location or stall type, the characteristics most relevant to relocation.

Table A3: Balance Table for Participants and Non-participants

	(1)	(2)	(3)
Variable	Non-Participants	Participants	Difference
Age	37.687	38.636	0.949
	(12.298)	(12.756)	(0.584)
Male	0.910	0.883	-0.027*
	(0.287)	(0.322)	(0.015)
Fresh Produce	0.495	0.592	0.097***
	(0.500)	(0.492)	(0.023)
Cooked Food/Drinks	0.505	0.408	-0.097***
	(0.500)	(0.492)	(0.023)
On Main Roads or streets	0.887	0.899	0.012
	(0.317)	(0.302)	(0.014)
On Pedestrian Walkways	0.113	0.101	-0.012
	(0.317)	(0.302)	(0.014)
Have Stalls/Carts	0.856	0.851	-0.005
	(0.351)	(0.356)	(0.016)
On the Floor	0.142	0.149	0.007
	(0.349)	(0.356)	(0.016)
Observations	564	2,922	3,486

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. The 3,486 observations are the 564 recruited vendors who did not complete the survey and the 2,922 of the analytical sample; the 13 who began the survey without finishing it appear in neither column.

A4 Covariate Coefficients

Table A4 reports the coefficients that Section 4.1.3 omits from the estimation table: the four design attributes held at the status quo in the main results, the elicited peer belief and the 27 vendor characteristics. The characteristics shift the mean of the vendor-specific valuation of a zone, and they drive the targeting results in Section 4.5.4. The indicator for missing education is perfectly predicted in 87 of the 200 bootstrap replications, where we hold it at the estimation bound. No representative-vendor quantity in the paper

uses it, so it affects no reported number, but it is not well identified and should not be interpreted.

Table A4: Vendor Characteristics

	(1) Linear	(2) Level dummies	(3) Main model (M0)
main			
Spots in the zone			
20	-0.031	-0.109	
	(0.056)	(0.066)	
40	-0.127*	-0.161**	
	(0.074)	(0.079)	
80	-0.041	-0.132*	
	(0.062)	(0.068)	
160	0.028	-0.038	
	(0.063)	(0.072)	
Expected competitors			
-10	-0.008	0.010	
	(0.085)	(0.084)	
-5	0.089	0.030	
	(0.057)	(0.062)	
+5	0.013	-0.045	
	(0.057)	(0.064)	
+10	-0.042	-0.130*	
	(0.072)	(0.077)	
Expected daily customers			
Lowest	-0.098	-0.145*	
	(0.075)	(0.076)	
Lower	-0.015	-0.112*	
	(0.068)	(0.066)	
Higher	0.000	-0.059	
	(0.078)	(0.077)	
Highest	0.044	-0.005	
	(0.069)	(0.070)	
Shock to the expected peer share			
-50%	0.088	0.028	
	(0.068)	(0.073)	
-20%	0.058	-0.016	
	(0.068)	(0.071)	

continued on the next page

Table A4: Vendor Characteristics *(continued)*

	(1) Linear	(2) Level dummies	(3) Main model (M0)
+20%	-0.031	-0.115	
	(0.068)	(0.071)	
+50%	0.120*	0.041	
	(0.063)	(0.068)	
Belief: peers relocating (%)	0.013***	0.013***	0.040***
	(0.001)	(0.001)	(0.004)
Peer belief missing	0.800**	0.805**	2.876**
	(0.404)	(0.405)	(1.388)
Spots (10s)			0.010
			(0.007)
Competitors (10s)			-0.160***
			(0.061)
Daily customers (10s)			0.113***
			(0.028)
Peers relocating (%)			-0.000
			(0.002)
Expected regulars following (%)	0.010***	0.010***	0.030***
	(0.002)	(0.002)	(0.006)
Expected volume: much fewer	-0.196	-0.194	-0.607
	(0.148)	(0.148)	(0.492)
Expected volume: slightly fewer	0.104	0.107	0.322
	(0.143)	(0.143)	(0.483)
Expected volume: about the same	0.171	0.169	0.531
	(0.183)	(0.182)	(0.594)
Expected volume: slightly more	0.392*	0.399*	1.134
	(0.234)	(0.235)	(0.751)
Expected volume: much more	0.518	0.519	1.434
	(0.315)	(0.317)	(0.916)
Regulars belief missing	1.559***	1.547***	4.058***
	(0.390)	(0.390)	(1.471)
Marketplace eviction frequency, leave-one-out	-0.034	-0.033	-0.118
	(0.056)	(0.056)	(0.190)
harassment = yes	0.171*	0.172*	0.481*
	(0.089)	(0.088)	(0.283)
eviction = yes	-0.003	-0.003	-0.001
	(0.084)	(0.084)	(0.267)

continued on the next page

Table A4: Vendor Characteristics *(continued)*

	(1) Linear	(2) Level dummies	(3) Main model (M0)
license = yes	0.300***	0.297***	1.013***
	(0.092)	(0.091)	(0.292)
log(weekly profit + 1)	-0.060	-0.060	-0.259
	(0.057)	(0.057)	(0.183)
Male	0.014	0.012	0.046
	(0.119)	(0.120)	(0.394)
Married	0.235**	0.235**	0.936***
	(0.098)	(0.098)	(0.326)
Number of children	-0.013	-0.014	-0.071
	(0.030)	(0.030)	(0.100)
Cognitive ability	-0.034	-0.035	-0.119
	(0.072)	(0.072)	(0.232)
Social ties (tea, information, family)	0.049	0.049	0.131
	(0.045)	(0.045)	(0.154)
Economic ties (minding, borrowing, lending)	-0.096**	-0.096**	-0.335**
	(0.046)	(0.046)	(0.151)
log(market size)	-0.110**	-0.112**	-0.310**
	(0.046)	(0.046)	(0.156)
Age (years)	0.002	0.002	0.006
	(0.003)	(0.003)	(0.011)
Education (years)	0.012	0.012	0.027
	(0.008)	(0.008)	(0.026)
Education missing	2.021**	2.048**	7.668***
	(0.921)	(0.928)	(2.875)
On a pedestrian walkway	-0.335**	-0.337**	-1.015**
	(0.134)	(0.135)	(0.431)
Away from main roads	0.253	0.250	0.981
	(0.250)	(0.249)	(0.890)
Pays an informal fee at the current spot	-0.132	-0.137	-0.454
	(0.149)	(0.150)	(0.465)
log(status quo daily customers)	0.103	0.103	0.386
	(0.083)	(0.083)	(0.288)
log(1 + status quo commute, km)	-0.030	-0.017	-0.146
	(0.076)	(0.079)	(0.248)
Observations	14550	14550	

Notes: The four held design attributes, the elicited peer belief, and the 27 vendor characteristics of the

main model. The attributes vary across choice occasions by design; the characteristics were all measured before the choice tasks. Standard errors clustered at the marketplace in parentheses. Columns match Table 5; the fixed-effects column is omitted because vendor characteristics are absorbed by the fixed effects. The first four blocks are the design attributes held at the status quo in the main results, entered as free level dummies in columns (1) and (2) and linearly in column (3); the base level is no change from the vendor's own status quo. The expected-volume rows are the categories of the pre-choice belief about customer volume at the new site, with the don't-know answer as the base. The two belief-missing rows flag item non-response on the belief questions, which are zero-filled so that no vendor is dropped. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A5 Balance for the Information Experiment

Table A5: Balance Table For Information Experiment

	(1)	(2)	(3)
Variable	Control Group	Treatment Group	Difference
Age	38.39	38.88	0.49
	(12.63)	(12.87)	(0.47)
Education	6.08	6.08	-0.00
	(4.77)	(4.75)	(0.18)
Male	0.88	0.88	0.00
	(0.32)	(0.32)	(0.01)
Married	0.82	0.82	-0.00
	(0.38)	(0.38)	(0.01)
Single	0.16	0.16	0.00
	(0.36)	(0.36)	(0.01)
Widowed/Divorced	0.02	0.02	0.00
	(0.14)	(0.15)	(0.01)
Backward Caste	0.88	0.86	-0.02
	(0.33)	(0.35)	(0.01)
Scheduled Caste/Tribe	0.09	0.10	0.01
	(0.29)	(0.31)	(0.01)
General	0.03	0.04	0.00
	(0.18)	(0.20)	(0.01)
Fresh Produce	0.60	0.59	-0.01
	(0.49)	(0.49)	(0.02)
Cooked Food/Drinks	0.40	0.41	0.01
	(0.49)	(0.49)	(0.02)
On Main Roads or streets	0.90	0.90	-0.00
	(0.30)	(0.30)	(0.01)
On Pedestrian Walkways	0.10	0.10	0.00
	(0.30)	(0.30)	(0.01)
Have Stalls/Carts	0.85	0.86	0.01
	(0.36)	(0.35)	(0.01)
On the Floor	0.15	0.14	-0.01
	(0.36)	(0.35)	(0.01)
Work 7 days a week	0.95	0.94	-0.00
	(0.22)	(0.23)	(0.01)
Hours of Operation	9.19	9.25	0.06

	(3.63)	(3.70)	(0.14)
Have employees	0.30	0.33	0.02
	(0.46)	(0.47)	(0.02)
Yesterday's Sales (INR)	2,927.72	3,039.17	111.45
	(2,477.57)	(3,211.32)	(106.40)
Yesterday's Profits (INR)	612.02	624.78	12.75
	(586.92)	(662.10)	(23.18)
Weekly Sales (INR)	18,493.02	20,014.04	1,521.03*
	(15,425.35)	(28,443.03)	(851.26)
Weekly Profits (INR)	4,270.38	4,249.35	-21.03
	(3,992.45)	(4,118.06)	(150.13)
Own Other Businesses	0.15	0.14	-0.02
	(0.36)	(0.34)	(0.01)
Never Changed Marketplaces	0.99	0.99	-0.00
	(0.08)	(0.08)	(0.00)
Upto 1 year	0.06	0.07	0.01
	(0.23)	(0.25)	(0.01)
1 to 5 years	0.21	0.21	-0.01
	(0.41)	(0.40)	(0.02)
5 to 10 years	0.21	0.19	-0.02
	(0.41)	(0.39)	(0.01)
More than 10 years	0.52	0.53	0.01
	(0.50)	(0.50)	(0.02)
Policy Awareness	0.08	0.07	-0.01
	(0.27)	(0.26)	(0.01)
Risk Aversion	3.57	3.62	0.05
	(1.72)	(1.73)	(0.06)
Observations	1,432	1,490	2,922

Note: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A6 Experimental Design Details

A6.1 Scenario Construction

We constructed the choice scenarios by filtering a full factorial design. Six attributes with five levels each give 15,625 combinations; for the peer share, the five levels are the shocks to each vendor's own belief. We applied two filters to keep the choice tasks realistic and cognitively manageable. First, we removed every scenario that weakly dominates the status quo, that is, every alternative at least as good as the status quo on all attributes and strictly better on at least one. Such scenarios reveal little about trade-offs. We retained "mixed" scenarios, in which some attributes improve while others worsen, along with the status quo scenario and scenarios in which only rent differs from current conditions. Second, we kept only scenarios in which at most three attributes differ from the status quo, so that respondents never weigh changes in all six attributes at once.

From the scenarios that pass both filters, we drew 150 at random with a fixed seed, so the draw is reproducible. We organised them into 25 blocks of six choice tasks each. We assigned each respondent to one block, so each vendor faced six binary choices between

a new market and the current one. Blocking balances statistical power across the sample against respondent fatigue. Random selection and assignment preserve variation in attribute levels without introducing systematic patterns.

Table A6 reports pairwise correlations between attribute levels in the design matrix. All are below 0.2 in absolute value, so the design supports separate estimation of attribute effects without multicollinearity concerns.

Table A6: Correlation Matrix of Experimental Design Attributes

	Market Rent	Number of Spots	Commuting Distance	Relocating Vendors	Competitors	Customers
Market Rent	1.00					
Number of Spots	-0.12	1.00				
Commuting Distance	-0.19	-0.08	1.00			
Relocating Vendors	0.08	-0.11	-0.15	1.00		
Competitors	0.11	-0.02	-0.11	-0.12	1.00	
Customers	-0.04	0.04	0.19	-0.12	0.05	1.00

Note: Correlations calculated using coded attribute levels (1-5) from the experimental design matrix.

A6.2 Bias Mitigation

Hypothetical bias. Two design choices limit hypothetical bias. We personalise the status quo to each vendor's verified current conditions (Taylor and Cummings 1999), and we tell vendors at the outset that their responses will inform actual PMC policy decisions, which gives them a reason to answer as they would act.

Inattention. The instrument includes a rent attention check. About 4.2 per cent of respondents failed it, so inattention is limited in our sample (Section 4.1.2).

Survey fatigue. Round dummies absorb any drift in responses across rounds, including drift from fatigue. Between choice modules, respondents also played a short incentivised game that measures risk aversion, which helps maintain engagement.

A6.3 Sample Stated-Preference Choice Task

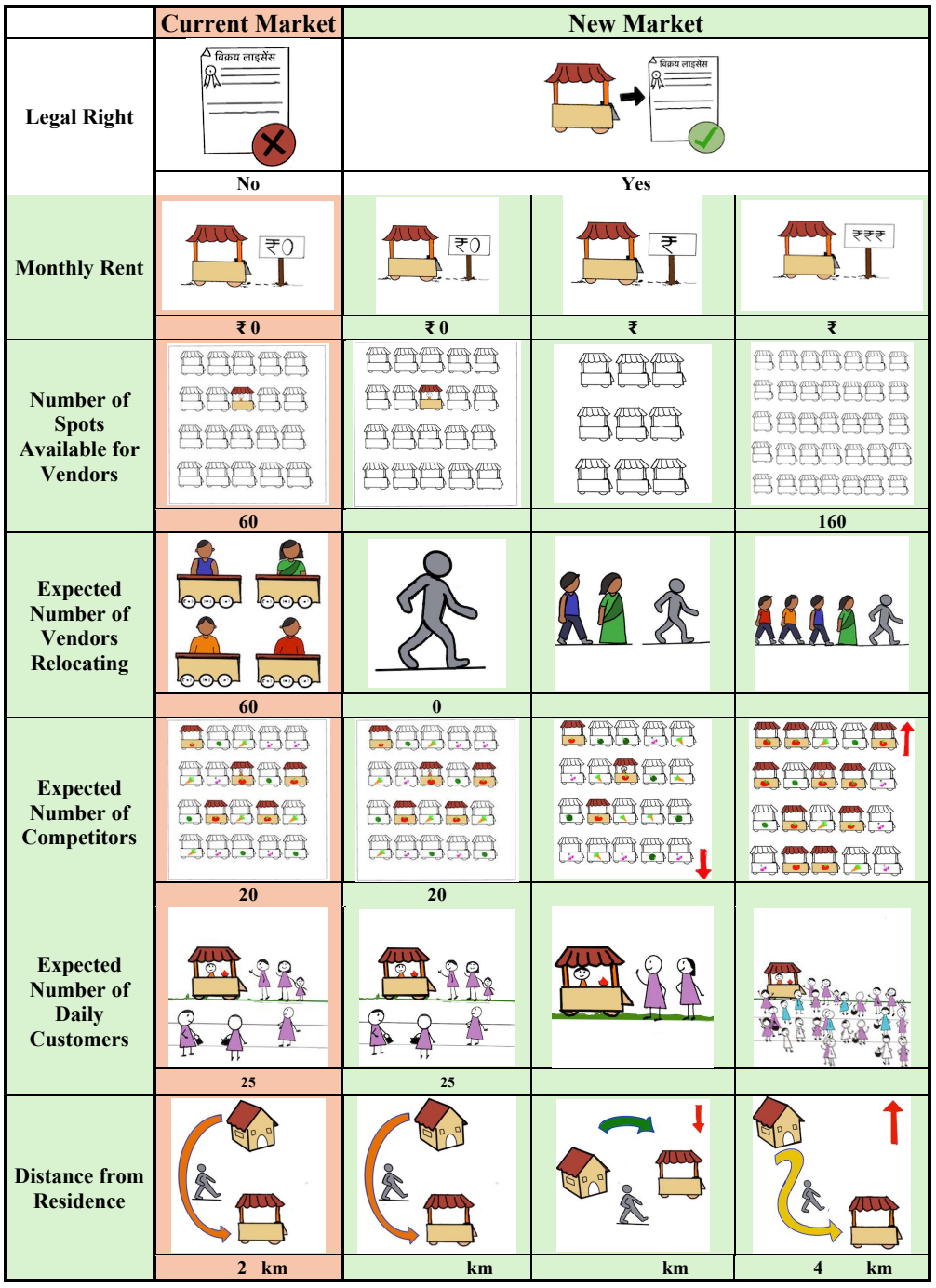
Figure A2 reproduces a sample choice task as presented to respondents. Because many vendors have low literacy, visual aids represented each attribute, and the enumerator read a verbal description aloud. The status quo column reflects each vendor's self-reported current conditions, elicited in the survey module immediately before the choice tasks.

Enumerator Script for Choice Tasks

Enumerators read the following script aloud before each choice task. The script was standardised across all enumerators and respondents.

> *For the rest of this exercise, we want you to imagine that you are being offered a stall in a newly established marketplace and you need to choose whether you*

Figure A2: Sample Stated-Preference Choice Task



Notes: The choice task presents two alternatives: the vendor's current informal location (left column) and a hypothetical formal vending zone (right column, with three sub-columns). Attribute values in the vending zone column vary across choice tasks according to the experimental design described in Section 3.1. The three sub-columns show the three possibilities for each attribute: the value stays at the status quo, decreases or increases. Visual aids replace text-based comparisons to accommodate low-literacy respondents. The practice round, which precedes the five analytical rounds, used the same format and is excluded from all estimation.

want to stay in the current marketplace or you want to move to this new market …

…The new market offered does not differ from your current market except for the characteristics listed here. Imagine that the amount of the products that you can store, and transportation costs for products remain the same in the new marketplace …

…In this exercise, we want you to imagine that each card represents a new scenario of the world. The only options available to you are the ones shown on the card. We want you to tell us which option you prefer out of the two. There are no "right" or "wrong" answers – you should only choose the option that you like better (even if you may not like any of them).

Table 2 describes the six attributes presented in each choice task and their possible values.

A6.4 Information Treatment Text

Control group respondents received no information and proceeded directly to round 4. Enumerators read the following aloud to treatment group vendors immediately after round 3:

Let me tell you about a policy that concerns businesses like yours. It is called the Street Vendors Act of 2014. Under this act businesses like yours are offered spaces in specified vending areas with formal stalls to sell products. This is a national level policy and after its complete implementation, street vendors will not be allowed to work in unauthorized vending areas and will face fines if found to do so. For example, in Chandigarh, the municipality has situated several vendors in new marketplaces. News in December 2024 reported that about 3,000 vendors who didn't abide by the policy are not being allowed to vend in their old location in Chandigarh. The vendors who choose to move to the new marketplace will not have the option to return to their old location once the policy has been fully implemented.

A7 Functional Form

Each attribute in the design takes five discrete levels, assigned at random, so each can enter utility as a full set of level dummies. That specification imposes no shape on any attribute and assumes only additivity across them. Every simpler form in the paper is a restriction of it, and we test each with a nested likelihood-ratio test. Table A7 reports the tests. The paper therefore has one maintained model and a sequence of testable restrictions on it, not a search across specifications.

Distance. On the full sample, the data reject a linear penalty above zero ($p = 0.041$). They do not reject restrictions (i), (iii) and (iv) in Table A7. The penalty for a commute one kilometre longer is 0.43 of the penalty for a commute three kilometres longer (95% CI [0.31, 0.56]). Linearity implies one third, which lies inside this interval. We retain the

free dummies, which nest every shape the offered levels can identify. We do not report a decomposition into a fixed cost of moving and a per-kilometre rate. The design offers few distinct distances, and the two terms move in opposite directions across samples while every reported quantity stays put, so the split is weakly identified.

Rent. The adopted form is a linear spline with knots at one and two per cent of sales. Against free rent dummies, it imposes one restriction, which the data do not reject ($p = 0.93$ to 0.95 across the two samples). The three segment slopes are 0.99, 1.84 and 0.64 kilometres of commute per percentage point. The data reject both a rent that carries no disutility below one per cent of sales ($p = 0.039$) and a rent that enters linearly from zero ($p = 0.008$).

Additivity. The maximum-rent ladder in Section 4.3 reads the cost of distance and the cost of rent off the same utility index, so it assumes the two are additive. This is the one maintained assumption the saturated model does not test. A fully interacted design is not feasible with 150 scenarios, and we did not construct the experiment to identify the interaction.

Table A7: Restriction Tests on the Saturated Model

Restriction	Attribute	χ^2	df	p	Verdict
Full sample, all five rounds					
(i) no step at zero	distance	0.01	1	0.9147	not rejected
(ii) linear above zero – D2	distance	4.18	1	0.0409	rejected
(iii) decrease effect is zero	distance	1.89	1	0.1695	not rejected
(iv) symmetry – D1	distance	0.39	1	0.5307	not rejected
(v) 1pct is free – R1 assumes	rent	4.25	1	0.0392	rejected
(vi) linear above 2pct – R3	rent	0.00	1	0.9488	not rejected
(vii) linear from zero	rent	11.81	3	0.0081	rejected
Marginal vendors					
(i) no step at zero	distance	7.31	1	0.0068	rejected
(ii) linear above zero – D2	distance	0.17	1	0.6793	not rejected
(iii) decrease effect is zero	distance	0.02	1	0.8864	not rejected
(iv) symmetry – D1	distance	8.11	1	0.0044	rejected
(v) 1pct is free – R1 assumes	rent	2.48	1	0.1152	not rejected
(vi) linear above 2pct – R3	rent	0.01	1	0.9297	not rejected
(vii) linear from zero	rent	29.11	3	0.0000	rejected

Notes: Each row is a nested likelihood-ratio test of one restriction against the free-dummy specification, which imposes no shape on any attribute. The adopted forms are those the data do not reject: free dummies for distance, and a linear spline in rent with knots at one and two per cent, whose single restriction against free rent dummies is row (vi). Rent being free below one per cent, assumed in an earlier draft of this paper, is rejected on the full sample, as is rent entering linearly from zero.

A8 Choice of Heterogeneity Model

Section 4.1.2 establishes that vendors differ. How to model that difference is a separate decision from functional form, and a harder one, because the candidate models are not all nested and no single test settles the choice. We hold attribute forms at the adopted specification throughout, so these models differ only in their treatment of heterogeneity.

Discrete types. A latent class logit with three classes estimates the share of each type, their utilities and class membership in one likelihood. It fits the distribution of choice counts very well, far better than any continuous specification, and its classes line up with a variable the model never saw.[6] The data reject four and five classes against three.

Continuous heterogeneity. A mixed logit with a normal random intercept beats the three-class model on log-likelihood, on BIC and on held-out prediction, and a Vuong test distinguishes them decisively ($V = 7.31$, $p < 10^{-12}$). We estimated two flexible alternatives to check whether the normal is the wrong shape: a non-parametric distribution estimated on a grid of support points with common slopes, and mixtures of two and of three normal components. Neither improves fit enough to justify its additional parameters. BIC values from models with different covariate sets are not comparable, so we do not report them side by side.

What the chosen model does not fit. The model we report predicts 273 vendors relocating in exactly one of the five rounds, against 199 observed, a χ^2 of 34.7 against the latent class model's 2.0. The flexible shapes above repair this but still lose on BIC. We report the model that predicts choices better out of sample, not the one that best reproduces this marginal distribution.

The form of the random distance response. The reported model lets the per-kilometre cost vary across vendors. One alternative lets the cost of changing site at all vary instead, and another lets both vary. Table A8 reports all five variants. Among the three forms with a single random distance coefficient, the per-kilometre slope fits best: by 53 BIC points against the any-move step and by 37 against the any-increase step. But one random distance coefficient is not enough. Adding a vendor-specific cost of any change in commute improves the likelihood by 11.3 log points for one parameter (LR 22.5, boundary $p = 1.0 \times 10^{-6}$) and BIC by 14.5 points. It is the only specification that also improves held-out prediction, on all three measures.

The richer model leaves the headline quantities unchanged. The break-even distance moves from 3.792 to 3.795 kilometres, the maximum rent at an unchanged commute from 4.48 to 4.44 per cent of sales, and aggregate predicted take-up agrees within 0.3 points at every design in Table 7. The intermediate rungs of the ladder move by at most

[6] The variable is the stated importance of legal rights, elicited after the choice tasks. The class that refuses the zone in every scenario is many times more likely to say legal rights do not matter at all. Because the item is elicited after the choices, part of this may be rationalisation.

eight per cent; the largest change is the maximum rent after a one-kilometre move, 2.09 against 2.25 per cent. The two models differ materially in the dispersion of the distance response over short moves, which the richer model more than doubles. They agree within about two per cent on who responds to a five-kilometre move and differ by a factor of 2.5 on a one-kilometre move. The quantities in Section 4.5.4 that depend on this dispersion, the responsive share and the take-up spread across vendors, are therefore lower bounds as reported. We report the simpler model because its standard errors rest on the bootstrap used throughout the paper. One assumption in this comparison is untested: we estimate the two random distance coefficients as independent. A correlated version would fit at least as well, so the statistics above are a lower bound on what the second coefficient is worth.

Table A8: Forms of the Random Distance Response

Random part	Log lik.	k	BIC	ΔBIC	Held-out	vs. M0
Per-kilometre slope only *(reported)*	-6005.87	53	12434.47	0.00	-1037.92	0.00
Any-move step only	-6032.40	53	12487.52	53.05	-1038.01	-0.08
Any-increase step only	-6024.42	53	12471.57	37.10	-1038.53	-0.60
Per-kilometre slope and any-move step	-5994.62	54	12419.93	-14.54	-1033.43	4.49
Per-kilometre slope and any-increase step	-5999.90	54	12430.49	-3.98	-1035.64	2.28

Notes: All five models share the adopted attribute forms and the same 27 covariates, and differ only in how the distance response varies across vendors. BIC is computed on vendors. Held-out is the log-likelihood on the control group's final two rounds, which no model saw. Of the two single random coefficients the per-kilometre slope is the better by 53 BIC points. Adding an any-move step to it improves the likelihood by 11.3 points for one parameter (LR 22.5, boundary $p = 1.0 \times 10^{-6}$), improves BIC by 14.5 and is the only specification that also improves held-out prediction. We report the simpler model and treat statements about dispersion at short distances as lower bounds.

A9 Within-Marketplace Ties

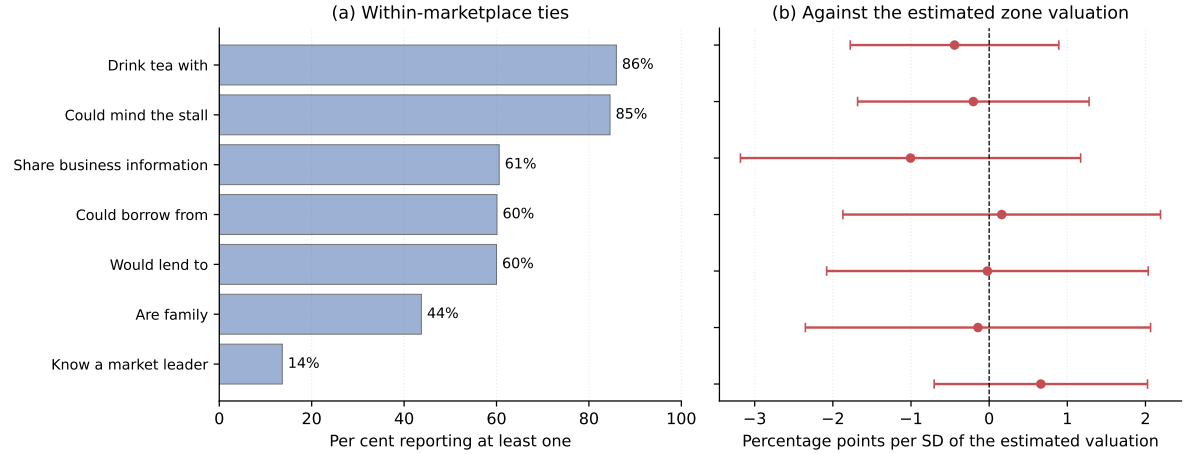
We asked vendors how many others in their marketplace they drink tea with, could ask to mind their stall, share business information with, could borrow from, would lend to, and are related to, and whether they know anyone in an informal leadership role. Panel (a) of Figure A3 reports the share with at least one tie of each kind. Social contact is widespread: 86 per cent name at least one vendor they drink tea with, and 85 per cent at least one who could mind their stall. Economic relations are less common and run equally in both directions: 60 per cent name at least one vendor they could borrow from, and 60 per cent at least one they would lend to. The median enumerated market has 50 vendors, so these counts are small relative to the market.

The counts record how many vendors a respondent holds each relation with, not which vendors. They are therefore within-marketplace degree counts, and no network statistic beyond degree can be recovered from them. They are also bounded by the current marketplace, so they miss ties elsewhere in the city and relations with customers or suppliers.

Panel (b) relates each tie to the estimated valuation of a zone. Each estimate comes from a separate regression of the tie indicator on the vendor's posterior valuation, with marketplace fixed effects and standard errors clustered at the marketplace. None of the

seven intervals excludes zero. The point estimates range from −1.0 to +0.7 percentage points per standard deviation of the valuation, against base rates between 14 and 86 per cent, so the null is tight, not uninformative. Ties within the marketplace do not predict how much a vendor values a zone. This is consistent with the null on the assigned peer share in Section 4.5.2, but it is a separate and weaker statement, because these counts cover only relations with other vendors in the current market.

Figure A3: Within-Marketplace Ties and the Estimated Valuation



Panel (a): share of the 2,910 estimation-sample vendors reporting at least one tie of each kind within their current marketplace, against a median enumerated market size of 50. Panel (b): one regression per relation of the indicator on the posterior zone valuation, marketplace fixed effects absorbed, standard errors clustered at the marketplace, 95 per cent intervals. The counts record how many vendors a respondent holds each relation with and not which ones, so no network statistic beyond degree is recoverable.

A10 Robustness

Is the distance cost status-quo bias? The paper attributes the distance cost to leaving the site, not to a general reluctance to depart from the status quo. Table A9 tests this directly. It adds to the adopted specification a step for *any* change from the vendor's status quo on each attribute, so each step is identified from a discontinuity at zero after the attribute's level is already in the model.

The step for distance is significant ($p = 0.026$). The steps for rent, expected competitors, expected customers and the assigned peer share are not. Rent is the informative case: it is the price-like attribute on which status-quo bias should bite hardest, and a vendor who simply disliked change would show a step there. The number of spots is the exception. It carries a significant step of -0.09 ($p = 0.039$), about two-thirds the size of the distance step in the same specification. The *level* of spots remains indistinguishable from zero, so vendors are indifferent to how many spots a zone offers but mildly averse to the number differing from their current market. This is a status-quo effect on one attribute, and we cannot rule out some aversion to change more generally. It is small, however, relative to a distance cost that consumes the entire value of the legal right within about four kilometres.

Table A9: Discontinuity at Zero, by Attribute

Attribute	Step	s.e.	p
Commuting distance	−0.134	0.060	0.026
Rent	−0.070	0.059	0.237
Number of spots	−0.093	0.045	0.039
Expected competitors	−0.026	0.050	0.605
Expected customers	−0.063	0.053	0.233
Assigned peer share	−0.005	0.051	0.918

Notes: Each row is the coefficient on an indicator for any change from the vendor's status quo on that attribute, entered alongside the attribute's own level. Standard errors clustered at the marketplace. Distance, rent and spots are estimated on a linear-in-level specification so that the step is separately identified: against the rent spline, "any rent change" is collinear with the first spline segment, and against the free distance dummies the step is collinear with the dummies themselves. The remaining rows are from the adopted specification with the steps added. The linear-in-level form understates the distance discontinuity relative to the free dummies, where the +1 km coefficient is −0.90 against a per-kilometre rate of −0.59; it is used here because a placebo requires the same functional form on both sides of the comparison.